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Experimental visual prosthesis

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ABSTRACT

Of the methods available which seek artificially to replace the sensory input lost through blindness, the approach which should ultimately be of greatest value is that by which visual information is fed directly to the brain. The paper reviews earlier work in this field and describes a new prosthesis which gives grounds for optimism.

1 INTRODUCTION

A prosthesis is an artificial device, such as a wooden leg, false teeth or an artificial hip, that replaces a part of the body which is absent or inoperative. A visual prosthesis is one that seeks to replace sight, in some measure, and it may perhaps be argued that the white cane, the ultrasonic-ray projector[1] the Optacon,[2,3] vibratory arrays[4] and many other devices which have been built or suggested to help the blind are all visual prostheses. In this paper, the kind of visual prosthesis to be considered is that outlined in Fig. 1. Here, the patient is represented by the contents of the broken-line box, and the normal visual pathway is, of course, by light from the scene to the eye, and by nervous transmission from the eye to the surface of the visual cortex of the brain, by way of the optic nerve and other relay structures which do not concern us here.

In patients in whom the eye or optic nerve has become inoperative through damage or disease, the prosthesis provides artificial vision by direct electrical stimulation of the visual cortex, patterned in space and time in a manner which seeks to emulate, as far as practicable, the normal neural cortical stimulation by nerve impulses arriving along the optic nerve. The prosthesis consists of two parts: an implant lying wholly within the body, which stimulates the cortex and which receives information, in the form of one or more radio-frequency signals, from an implant driver outside the body. Information is fed to the implant driver from some suitable source, most obviously a camera, but alternatively a storage medium such as punched paper tape, or from an array of manually operated switches.

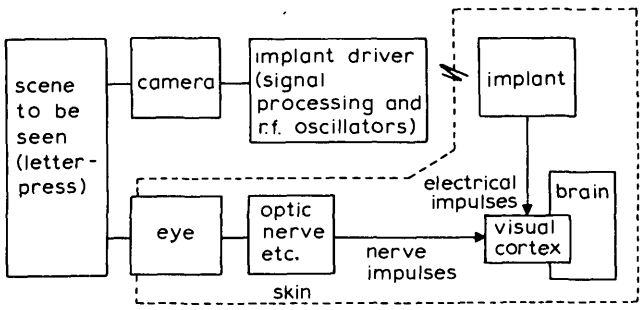

Fig. 1

Principle of visual prosthesis

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The first visual prosthetic device was built by Brindley, and was implanted into a patient in July 1967.[5] The driver consisted of a battery of pulsed valve oscillators (crosscoupled ELL 80s) operating at frequencies in the range 5-10 MHz, whose inductors were in the form of probes 9·5 mm in diameter, which could be hand-held or supported in a helmet-like jig. The implant consisted of two main parts:

(a) an extracranial part, a silicon rubber shell lying between the patient's scalp and her skull, containing 80 very small epoxy-resin-encapsulated 'radio' receivers, each having the circuit shown in Fig. 2. These pick up the induction field from the driver probes

(b) an intracranial part, a silicon rubber shell lying closely over the visual cortex of the patient's right cerebral hemisphere, and bearing 80 small (1 mm^2) platinum stimulating electrodes.

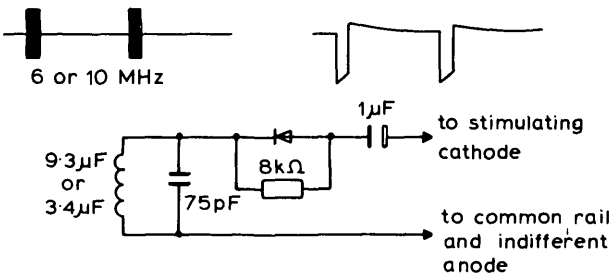

Fig. 2

Circuit of one radio receiver in Brindley's first implant

The implant contained 80 such receivers

These two parts were joined by a short length of flexible 80-core cable which passed through a small hole at the back of the skull. A return path for stimulating currents was provided by a number of large platinum indifferent electrodes attached to the outer surface of the extracranial part. The equipment demonstrated the promise of this type of visual prosthesis. When a driver probe of appropriate frequency was held against the patient's head so as to oppose one of the receivers in the implant, the patient would report a phosphene, or perception of light, whose brightness depended on the stimulus strength, in a definite fixed point in the visual field. Many phosphenes were small white points of light 'like a star in the sky'. Others were more diffuse, or elongated in shape 'like a matchstick'. By relating the apparent position of a phosphene to the cortical position of the electrode eliciting it, it became possible to confirm in greater detail the neuropathological evidence, from patients suffering brain damage, connecting neuroanatomy and visual function. By the simultaneous use of several driver probes, a number of phosphenes could be combined to form simple patterns. In Fig. 3, the patient may be seen describing such a pattern.

Encouraged by these results, we have now developed an improved visual prosthesis which was implanted on the 4th February 1972. In gross mechanical terms, the new implant is similar to the first, in that it has an extracranial part containing microelectronics, an intracranial part carrying stimulating electrodes, and cables between them; it differs in that there are two stimulating arrays, one for each hemisphere. Electrically, the new prosthesis is rather different from the old; moreover, the materials used in constructing the implant are different. The engineering aspects of this new work are described in this paper.

Fig. 3

Prof. Brindley testing first patient to receive visual-prosthetic implant

2 THEORETICAL SYSTEM DESIGN

Brindley's first implant, with its 80 separate channels, has one important advantage and one important disadvantage over alternative systems that can be considered. The advantage is that implants built in this way are relatively immune to faults developing inside them; a short circuit or a broken wire typically 'loses' one channel, which is unfortunate but not disastrous. The disadvantage is that the patient has hair which varies in length, and in how it is combed, from time to time. It is difficult to device a 'helmet', which the patient can put on himself, so that 80 oscillator probes always come opposite their respective receivers. The scalp is not a very rigid structure, and a displacement of the helmet of 5 mm from the intended position is liable to cause severe crosstalk between channels, although the situation can be improved by tuning adjacent receivers to different frequencies. Evidently, it is easier to tolerate misplacement of the helmet if one can arrange to have fewer r.f. inductive links further apart. Yet, to make a better prosthesis, one needs more channels, and so more phosphenes to choose from. It is difficult to see much hope of achieving this at the rate of one r.f. link per phosphene.

At the opposite extreme, one can imagine a system having only one radio-frequency channel, and that insensitive to probe position, at least over a range. Using stimulus pulses not more than 50 μs in duration and a stimulus repetition time of 10 ms at any given electrode, it is clear that pulses can be distributed in succession to up to 200 stimulating electrodes, from a single receiver, by way of gates opened and closed by a ring counter. Immunity to probe-position variations might be sought by using pulse-duration instead of pulse-amplitude modulation, and placing a limiter across the output of the receiver. Such a proposal would be convenient for the user if the implant worked perfectly, but a fault anywhere in the ring counter would make the whole implant useless.

Marg et al.[6] have proposed a visual-prosthetic implant controlled through only four r.f. channels. Their system offers random access to the electrodes; useful electrodes can be used in any order, and no time need be lost addressing useless electrodes, as would be the case with the ring-counter proposal. There are vulnerable points within the implant where the effect of a single fault could be serious. Evidently, the authors are aware of this, and mention the possibility of introducing triple redundancy into the implant circuitry, though they do not discuss how this would be implemented.

Clearly, there is a tradeoff between, on the one hand, the degree of 'fanout' between the number of r.f. channels and the number of electrodes, and, on the other, the susceptibility of the implant to failure caused by internal faults. In 1967, Brindley devised the 'row-and-column-matrix' implant.[7,8] If one imagines a plane matrix having m rows and n columns, a radio receiver associated with each row and each column, and, at each intersection between a row and a column, a 2-input AND unit feeding a stimulating electrode, electrode i, j will only be excited ($0 < i < m$, $0 < j < n$) if row receiver i and column unit j are simultaneously excited. In this manner, $m \times n$ electrodes are controlled by $m + n$ radio channels. An obvious difficulty with this proposal arises when one seeks to excite two or more electrodes together; for example, to stimulate electrodes 2, 1 and 5, 5, one excites row receivers 2 and 4, and column receivers 1 and 5. Evidently, this will also excite electrodes 2, 5 and 4, 5. The difficulty is easily evaded by 'scanning the rows' (or columns). Fortunately, the stimulus pulses to be delivered are brief compared with the interval between them. Hence the row and column whose simultaneous excitation operates a given intersection, delivering a stimulus pulse therefrom, need themselves only be 'on' for a small proportion of the time. If the system is timed so that the rows (or columns) are used sequentially, it will work properly for any combination of electrodes required. Since it is not necessary, for the patient to perceive two phosphenes at once, for the stimulus pulses which produce them to occur simultaneously [i.e. synchronous trains of nonsynchronous pulses are as effective as synchronous trains of synchronous pulses (Fig. 4)], the need to scan the rows is of no consequence.

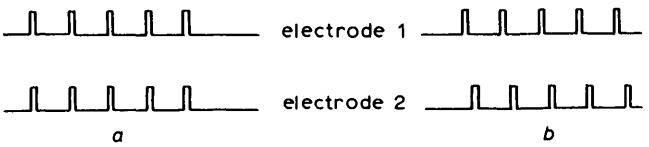

Fig. 4

Synchronous trains of pulses

a Synchronous pulses
b Asynchronous pulses
Both will cause a pair of phosphenes to be seen for the duration of the train

The row-and-column notion may be extended to matrices of more than two dimensions. The fanout available is evidently greatest if plane matrices are made square, cuboidal matrices are made cubical, etc. For a p-dimensional hypercubic matrix containing Q intersections, the number of points along each edge will be $\sqrt[p]{Q}$, and the total number n of rows and columns, i.e. the total number of radio channels, will be $p\sqrt[p]{Q}$. The broken straight lines in Fig. 5 represent n as a function of Q for various p. Evidently, the use of a high-order matrix gives a useful economy in the number of radio channels when the number of intersections is very large (10^4), offers little advantage when the number of intersections

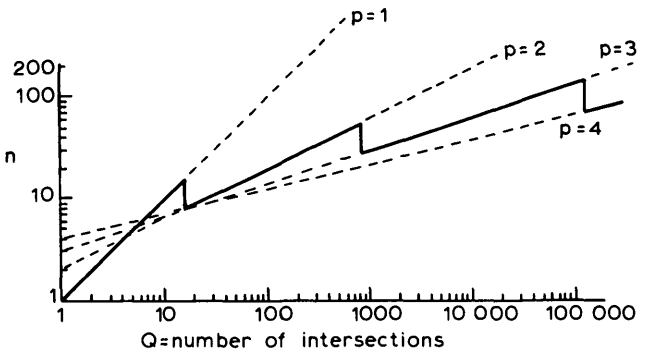

Fig. 5

Design guide for matrix implants, suggesting correct order of matrix for a given number of intersections $n = p\sqrt[p]{Q}$

is about 100, and is actually disadvantageous when the number of intersections is below 10. By choosing an arbitrary switching criterion, such as that, as the number of intersections is increased, one will move up to a higher order of matrix when to do so offers a halving in the number of radio channels, one arrives at the solid line in Fig. 5. This may be used as a design guide for matrix implants; it suggests that a 2-dimensional matrix is appropriate where the number of intersections lies between 16 and 800, and a 3-dimensional matrix if the number is between 800 and 100 000. We have thought entirely in terms of 2-dimensional matrices, and implants having between 50 and 500 intersections.

3 PRACTICAL DESIGN OF IMPLANT

3.1 General form

The fact that the material used for the extracranial shell of the implant should be compatible with tissue, castable and suitable for sterilisation in an autoclave suggests that it should be made of, or at least be encased in, silicon rubber. Surgical considerations dictate that the shell should have a maximum arc length, measured from front to back, of about 180 mm, and a maximum arc width, measured from side to side of, perhaps, 125 mm. Surgical and cosmetic considerations require the thickness to be not more than 6·5 mm at the centre, tapering away gradually to zero at the periphery. If the implant is to fit a typical skull, these dimensions imply considerable curvature of the implant in both directions. Hence, the microelectronics of the implant cannot be made using conventional procedures, which are essentially plane, or at best cylindrical, in one piece. One is therefore led to the notion of a number of small, flat, rigid packages disposed throughout the flexible body of the shell, and interconnected by insulated flexible wires. Fortunately, the circuitry of a row-and-column implant can be divided conveniently into packages of a convenient size, neither so small that an enormous amount of inter-package wiring is needed, nor so large that the packages will not fit into a shell having the degree of curvature required.

3.2 Choice of technology

At an early stage, we had to choose the microelectronics technology by which the packages should be made. There seemed to be four options open to us:

(a) miniature conventional components
(b) thin-film hybrid circuitry
(c) thick-film hybrid circuitry
(d) silicon monolithic circuitry.

The advantages and disadvantages of these methods, as seen by us in 1968 and shown in Table 1, seemed to weigh in favour of the use of thick-film hybrid circuitry, and although today we might alter some of the judgments made, we have no reason to regret our decision to use thick-film-circuit methods.

TABLE 1

FACTORS AFFECTING CHOICE OF MICROELECTRONICS TECHNIQUES

Technique	Capital cost of facility	Device cost	Component-packing density	Voltage and power capability	Range of resistance available	Electrical precision	Suitability for autoclaving
Miniature conventional components	low	low	moderate	good	wide	good	possibly
Thin-film hybrid circuit	highish	moderate	high	good	low values only	excellent	yes
Thick-film hybrid circuit	moderate	moderate	high	excellent	wide	good	yes
Silicon monolithic integrated circuit	very high	very high except in huge quantities	very high	poor	low values only	poor	yes

3.3 Choice of packaging

The medical grade of castable silicon rubber which we wished to use for the implant body has, in common with other silicon rubbers, a high water-vapour transmission rate. Further, it does not normally form a bond with other materials. Hence, in the moist environment in which the implant has to work, vapour will pass through the bulk of the rubber and will condense in the small voids between the rubber and any electronics components contained in it, covering such components with a thin film of water. It is therefore no use expecting the implant body to keep the microelectronics contents dry.

There are at least three ways by which implanted microelectronics may be affected by any water, and possibly salt, which reaches it:

(a) Some components may be inherently affected by water vapour, sodium ions, chloride ions etc. and may cease to work.
(b) The presence of a water film over the surface of components may provide a shunt conduction path which, at the very least, will modify their apparent electrical performance.
(c) As a consequence of current flow through water films, metals may become corroded as electrolysis takes place, and some very reactive products may be liberated, for example H_2, O_2, Cl_2 and NaOH.

Components which are susceptible to water vapour etc. must be protected by putting them in hermetic enclosures. Nothing less is acceptable, since all plastics are porous, and a water-vapour-susceptible component encapsulated in a plastics, rubbery or resinous material will eventually fail, however low the vapour-transmission rate of the encapsulant may be.

The development of a water film—and hence of spurious conduction paths and electrolysis—can be prevented if a waterproof adhesive sealant can be found which forms a bond to the components within it and which does not 'let go' in the presence of water vapour. Such a sealant will then have three uses:

(a) as an encapsulant for components which are not susceptible to water vapour
(b) for sealing the connections to such components
(c) for sealing the connections to any hermetic enclosures used.

There must be joins between the flexible insulated wires used to wire up the implant and the metal of the metal-in-glass seals of the hermetic enclosures; it will be necessary for the sealant to insulate these joins.

We therefore began the development of the new implant by setting up three long-term test programmes. The first was to check for water-vapour susceptibility all the types of hybrid thick-film component we expected to use; the second was to find a satisfactory sealant. Strictly, we should have completed the sealant tests first, as one cannot measure the effect of water vapour on an encapsulated component submerged in saline until one has excluded the possibility that any change of performance encountered is attributable to a surface water film. In fact, we began the vapour-susceptibility tests first, thinking that we already had a satisfactory sealant in the room-temperature-curing expoxy resins. Such resins are successfully used in implantable cardiac pace-

makers, and we hoped that they would prove satisfactory for use. Eventually, we found sealants which were better—at least in our hands—than epoxy resin, but the results of the vapour susceptibility tests, using epoxy resins as sealant, were sufficiently unequivocal to enable us to proceed. The third set of tests was to find a satisfactory insulation for the wires for connecting between package and package, and between the packages and the stimulating electrodes. The leads to the electrodes form the output cables and the insulation here is immersed in body fluid. It is essential that the insulation remain perfect under this condition.

3.4 Water-vapour-susceptibility tests

We tested the following types* of thick-film hybrid component for susceptibility to water vapour:

150 pF medium-relative-permittivity ceramic-chip capacitor
10 000 pF high-relative-permittivity ceramic-chip capacitor
'Certified' thick-film resistor
passivated microdiode
planar-transistor chip from various manufacturers
1 μF 50 V tantalum capacitor, nonhermetic
1 μF 50 V tantalum capacitor, hermetic
55 V plastics-encapsulated Zener diodes

Results of these tests were as follows.

3.4.1 150 pF chip capacitors

Two capacitors had a length of polyvinyl-chloride-coated tinned-copper flexible wire soldered to each termination pad. The assemblies were defluxed, and then each was drawn into a short glass tube so that one wire emerged from each end of the tube. The tube was then filled with a warmed commercial epoxy-resin adhesive† and, when the adhesive had cured, the assembly was lowered into a bath of 1% saline kept at 50°C (Fig. 6). The leads emerged through the surface and were connected through a switching arrangement, to a 1000 Hz a.c. bridge, enabling either capacitor to be checked from time to time. The results are shown in Fig. 7. The capacitance of one specimen was extremely steady, with negligible tan δ. The capacitance of the other was less steady, and the tan δ rose to 0·25 during the first six months, then declined again. A tan δ of 0·25 in a 150 pF capacitor at 1000 Hz is not as serious as may appear at first sight, since it corresponds to a shunt leakage of several megohms, 1000 times greater than the impedance of the circuit in which we intended to capacitor should be used.

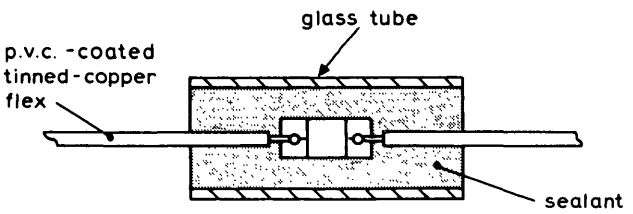

Fig. 6

Method of preparing 2-terminal component (e.g. chip capacitor) for testing while submerged in saline

3.4.2 10 000 pF chip capacitors

Two capacitors were prepared for testing, and were submerged as in Fig. 6. Over 15 months, the capacitance of both specimens drifted down to about 90% of the initial value and then showed signs of levelling off. Tan δ remained low after an inexplicably high initial value (Fig. 7a).

3.4.3 Thick-film resistor

Polyvinyl-chloride-insulated wires were soldered to the palladium-silver resistor terminations. After being defluxed, the substrate carrying the resistor, its terminations and wire connections was placed in a shallow glass-fibre-reinforced (g.f.r.) diallyl-phthalate tray, and the tray was then filled with the adhesive (Fig. 8). When the adhesive had cured, the whole assembly was submerged. Over 15 months, the resistance measured on the 1000 Hz bridge drifted down to 87% of its initial value; most of the change occurred in the first six months (Fig. 7b).

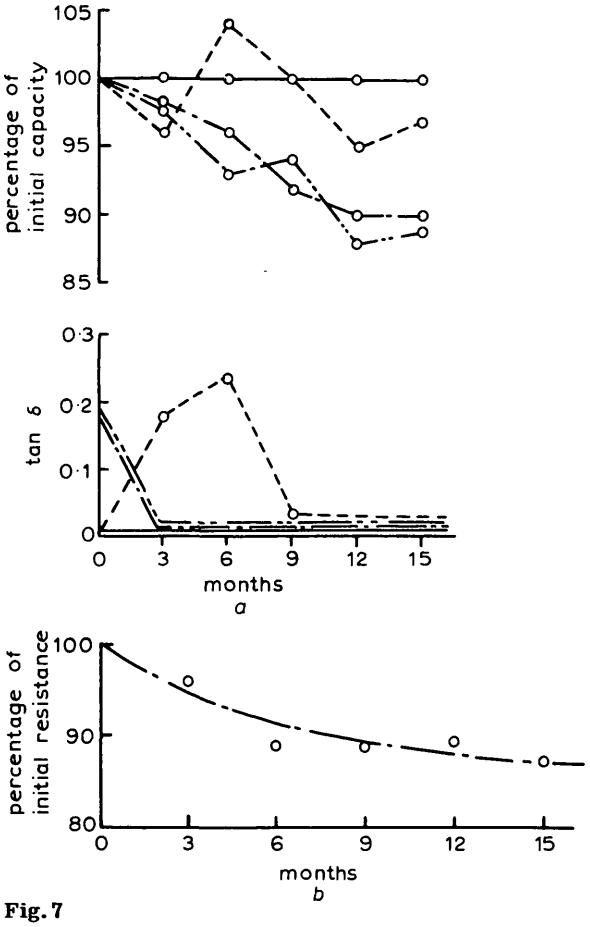

Fig. 7

Results of tests for submerged components

a Ceramic chip capacitors
—— *150 pF*
----- *medium relative permittivity*
— — *10 nF*
—·—·— *high relative permittivity*

b Thick-film resistor

3.4.4 Passivated microdiodes

Three samples were prepared as shown in Fig. 6, two sealed in the adhesive used previously and one in a different compound,‡ and were then submerged. The leads were taken via a selector switch to a suitable transistor tester so that the performance of any diode could be checked from time to time. After two years, all three diodes had developed a reverse leakage which did not exceed 100 μA at 100 V, quite acceptable for our purposes.

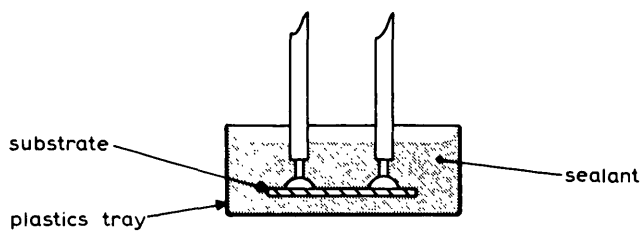

Fig. 8

Method of preparing thick-film circuits or patterns for testing while submerged

* *Supplied by Erie, Du Pont, TRW Semiconductors, General Electric, Union Carbide, Motorola, Mullard and others*

† *Domestic Araldite*

‡ *Araldite 752/956*

3.4.5 Planar transistors

74 nonhermetic transistors were tested for susceptibility to water vapour, encapsulated either in wax* or in adhesive silicon rubber† or in epoxy resin.‡ These three groups of materials were chosen as the most promising after preliminary experiments with a much wider range of likely encapsulants, which included beeswax/paraffin-wax mixtures, epicien wax, photoresist and sulphur. The transfer-moulded epoxy-encapsulated samples§ were prepared for submersion by soldering polyvinyl-chloride-insulated tinned-copper flexible wires to the transistor leads, placing a short piece of glass tube around the soldered joints, and filling the tube with sealant‖ (Fig. 9). The DK9 epoxy-encapsulated samples, and some early wax-encapsulated samples, were tested with the dice mounted on TO18 headers (Fig. 10), and with an approximate hemisphere of the test encapsulant over the die. Subsequently, we realised that this method of testing might give ambiguous results, since one cannot be sure that a transistor that fails when encapsulated in this way has done so because of diffusion of vapour through the bulk of the encapsulant; it might have failed from failure of the adhesion between the encapsulant and the header, with the consequent appearance of a water film on the die. We therefore adopted the construction shown in Fig. 8, in which the transistor dice were bonded to small thick-film substrates, and had their connections brought out to printed palladium-silver terminations. The encapsulant also acts as the sealant.

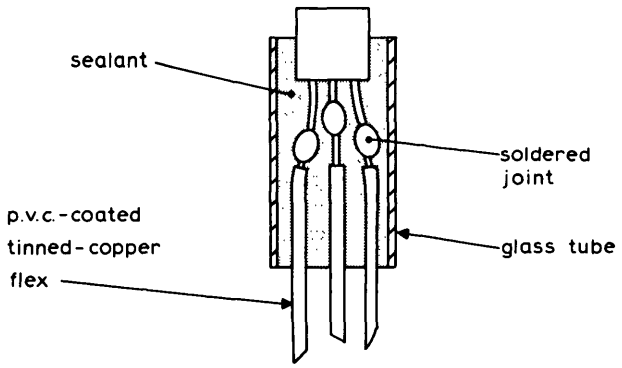

Fig. 9

Method of preparing professionally encapsulated transistors for testing while submerged

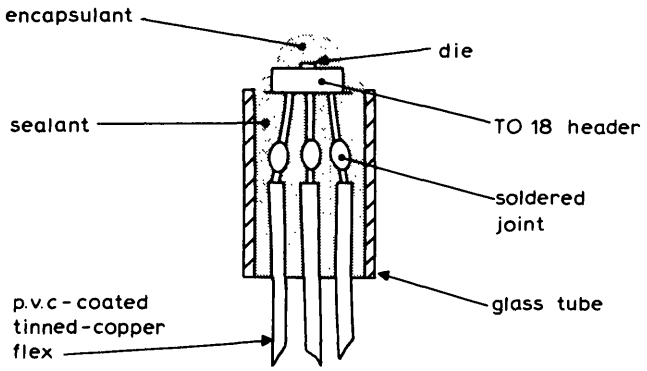

Fig. 10

Method of preparing experimentally encapsulated transistors for testing while submerged

All the silicon-rubber-encapsulated, room-temperature-cure epoxy-encapsulated, and the later wax-encapsulated samples were prepared in this way.

After submersion, the wires from all transistors were taken through a selector switch to the transistor tester, so that the performance could be checked from time to time. No bias was applied to the junctions except when the transistors were actually under test. Test voltages up to 60 V were used.

The results of these tests are shown in Fig. 11. Two types of transistor failure were most commonly observed, irrespective of whether the transistors were n-p-n (mostly BC 108) or p-n-p (2N 2904): a decline in V_{ceo} or a rise in I_{ceo} (or both). β was little affected. Occasionally, the device would become open-circuited or short-circuited.

The Figure shows that it is rare for a nonhermetic transistor to survive submersion for more than three months, although occasional specimens do much better than this; it also shows that silicon rubber is worse than epoxy resin as an encapsulant, while wax is perhaps a bit better. This ranking is in agreement with what one should expect from consideration of the respective water-vapour-transmission rates.

It may be objected that the noncatastrophic failures (decline in V_{ceo} and increase in I_{ceo}) observed in these specimens need not have been caused by any harmful effects of water vapour on the silicon chip, but could have been caused merely by a sealant failure. This is because water-film leakage from collector to base would simulate an increase in I_{ceo}, and a further leakage from base to emitter would convert the transistor into a 'variable Zener type', simulating a decline in V_{ceo}. This argument may be demolished by demonstrating that hermetic transistors, sealed in a similar way and with similar materials to the non-hermetic transistor, do not fail, or at least do not fail so quickly.

44 hermetic transistors (2N 2220 in TO18 cans) were therefore prepared in a similar manner to that shown for the transistors in Fig. 9, were submerged, and were connected to the selector switch so that their progress too could be monitored on the transistor tester. Fig. 12 shows the results, and clearly the hermetic specimens do last longer. A quite unlooked-for result of this experiment was the hint that adhesive silicon rubbers might be better sealants than epoxy resins. Specimens in silicon rubber which failed quickly were those from early batches, before we had much experience in preparing surfaces for bonding to adhesives. No failures have occurred in submerged hermetic transistors, sealed in adhesive silicon rubber, from our later batches.

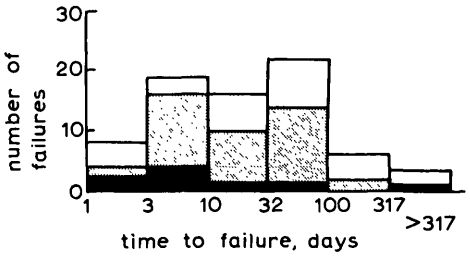

Fig. 11

Results of tests on 74 nonhermetic transistors

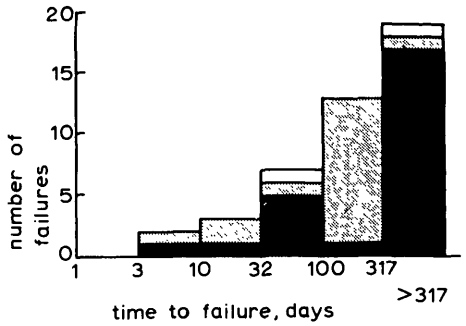

Fig. 12

Results of tests on 44 hermetic transistors

silicon rubber

epoxy resin

wax

* *Kel F chlorotrifluoroethylene wax*
† *Either Midland Silicones Silastoseal E or equal volumes of xylene and Dow Corning Medical Adhesive type A*
‡ *Either transfer-moulded or Araldite room-temperature cure or Hysol DK9 high-temperature cure*
§ *T I 'Silect'*
‖ *Using Araldite 752/956*

3.4.6 Nonhermetic tantalum electrolytic capacitors

Nonhermetic (i.e. plastics-sealed) tantalum capacitors are much smaller than hermetic components of the same capacitance and voltage rating; so it is useful to know how applicable they are for implant use. Our evidence here is conflicting. A first batch of four 1 μF 50 V specimens, prepared as in Fig. 6 and sealed in epoxy-resin adhesive,* was submerged and connected to the 1000 Hz bridge via a selector switch and 4·5 V polarising battery. When a capacitor was not actually being measured, no polarising voltage was present. After a year, no perceptible changes had occurred in the measured capacitance of the batch, but the breakdown voltage was found to have fallen to below 10 V. A second batch of three capacitors was then prepared, sealed in adhesive silicon rubber and arranged so that the transistor tester could be used to apply a rectified sinewave of 50 V average value from time to time, to monitor the ripple current and to yield a crude estimate of the capacitance. Tested in this way, none of the three failed catastrophically in nine months, but the ripple currents are not very steady. The capacitance appears to be maintained.

It is possible that the failure of the first batch of capacitors, and the nonfailure of the second, is attributable to the use of different sealants, or it may be caused by the difference between the tests; this question remains to be investigated.

3.4.7 Hermetic tantalum electrolytic capacitors

One 1 μF 50 V hermetic capacitor was prepared as in Fig. 6, sealed in adhesive silicon rubber, submerged and connected so that it could be checked from time to time on the transistor tester, in the same manner as the nonhermetic samples in Section 3.4.6. In nine months, no degradation has been observed; the ripple current is very steady, and the capacitance and breakdown voltage are maintained.

3.4.8 Plastics-encapsulated Zener diodes

Two 55 V Zener diodes in DO7 enclosures were prepared as in Fig. 6, sealed in epoxy-resin adhesive,* submerged and connected so that their performance could be checked from time to time on the transistor tester. No change in performance has been detectable in two years. Presumably, this implies that the junctions are carefully passivated.

3.4.9 Conclusions from water-vapour-susceptibility tests

Our conclusions from the foregoing tests were as follows. It may be possible to use unpassivated planar transistor chips in implant construction, but it will almost certainly be unprofitable to try to find out how to do it. Transistor chips should be used in hermetic enclosures, or alternatively passivated chips may be satisfactory. In contrast, the changes in performance observed in chip capacitors, microdiodes and the thick-film resistor are probably due less to inherent changes in the components themselves than to deficiencies in our knowledge of the correct choice and method of use of sealants. These components can probably be used without hermetic enclosure, as can Zener diodes and hermetic tantalum capacitors, particularly if better sealants can be found.

3.5 Search for a sealant

The requirements of a sealant are:

(a) It must form an adhesive bond to the surface of the electronics components, substrates, soldered joints etc., it is to protect, and this bond must not weaken with the passage of time in the presence of water.

(b) It must slump in a time short compared with its curing time, so that all crevices are penetrated by the sealant and all air is driven out. In practice, this seems to imply a viscosity of about 20 Pa s.

(c) It should have a low vapour pressure, so that, if one decides to help the processes of crevice penetration and air removal by pouring at reduced ambient pressure, one is not frustrated by 'boiling' of the sealant.

(d) It must not shrink on curing. It follows that one may not reduce the viscosity of a stiff material by the addition of solvents.

(e) It should not evolve corrosive or toxic reaction products in curing.

(f) Since it is used in many applications of small quantities, it is convenient if the sealant is of the 'one-pack' kind and has good 'shelf' life.

As was explained in Section 3.3, we began by using epoxy resins for sealing because there seemed to be good precedent for it. At the same time, we realised that epoxy resins might not prove satisfactory because the requirements in a visual prosthetic implant are different from those in a cardiac pacemaker in two important respects:

(a) The electric-field strength the sealant will have to withstand. A pacemaker typically has a maximum potential difference present of 6 V, and wires with this potential difference need not come within less than, say, 1 cm of each other, producing a field strength of 0·6 V/mm. Experience with the first visual implant suggested that a maximum output-pulse amplitude of 60 V to any combination of stimulating electrodes was desirable in the second implant, to have ample stimulating capability in hand. Using 250 μm (0·01 in nominal) spacing between adjacent microcircuit-package-connection lands, the electric field could reach 240 V/mm, 400 times that in the pacemaker

(b) The implant should withstand sterilisation in an autoclave at 130°C. This could eliminate a number of otherwise promising epoxy materials. When the hermetic-transistor vapour-susceptibility tests suggested that adhesive silicon rubber might be a better sealant than expoxy resin, it became imperative to find out whether this was so. Silicon rubbers are stable at 130°C, so that the autoclavability problem would have been solved as well.

A special test pattern was therefore devised (Fig. 13) which could be printed on to 25·4 mm-square (1 in nominal) thick-film substrates. A variety of conductor inks were used, containing different metals or combinations of metals, so that any effect the choice of metal might have in determining the effectiveness of the sealant under investigation would be revealed. Insulated flexible wires A, B, C and D (platinum wires insulated with etched p.t.f.e., see Section 3.6) were soldered to the four corner lands. Assemblies were then defluxed and each suspended in a g.f.r. diallyl-phthalate tray, not touching the bottom or sides, and sealant was poured in until the tray was full (Fig. 8). When the sealant was cured, the assemblies could be submerged and connected up to the simple testing circuit shown in Fig. 13.

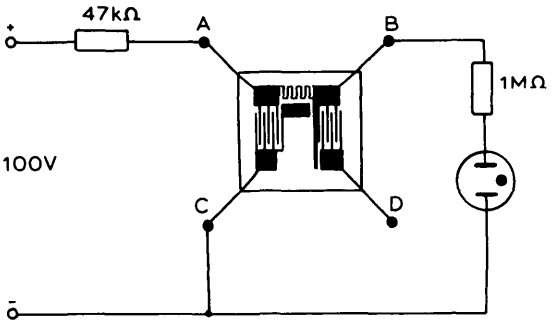

Fig. 13

Test pattern printed in conductor ink on thick-film substrate with ancillary circuitry

Spacing between adjacent electrodes in interdigitated 'field-strength cells' is 250 μm

Should a water film develop between the sealant and the test pattern, two failure modes are to be expected:

(a) electrolytic destruction of the sinuous anode by current flowing to the large rectangular cathode beneath it

(b) development of a short circuit somewhere in the 'field-strength-test cell' on the left, where the field strength is 200 V/mm, as a result of first a leakage current, and secondly the accumulation of electrolytic debris.

In either case, the neon warning lamp will go out. The bottom right-hand wire D runs to a control test cell, where the corresponding decline in insulation resistance not in the presence of the electric field can be monitored.

Fig. 14 shows the results of the experiment on the day on which a batch of three submerged samples successfully reached 500 days. The most obvious conclusion to be drawn

* *Araldite 752/956*

is that the adhesive silicon rubbers are, indeed, the most reliable sealants tried.*

So far as could be determined, all samples which failed did so because of insulation breakdown in the 'field-strength cell'. The results do not support the idea that submerged equipment sealed in short-lived materials will last longer if noble metals are used for the conductors. It is true that, in both epoxy resin and wax, the tinned-palladium-silver samples, for example, failed sooner than the platinum-gold samples, but the difference, even if significant, is too small to be considered important. We conclude that the choice of metal is uncritical compared with the correct choice of sealant.

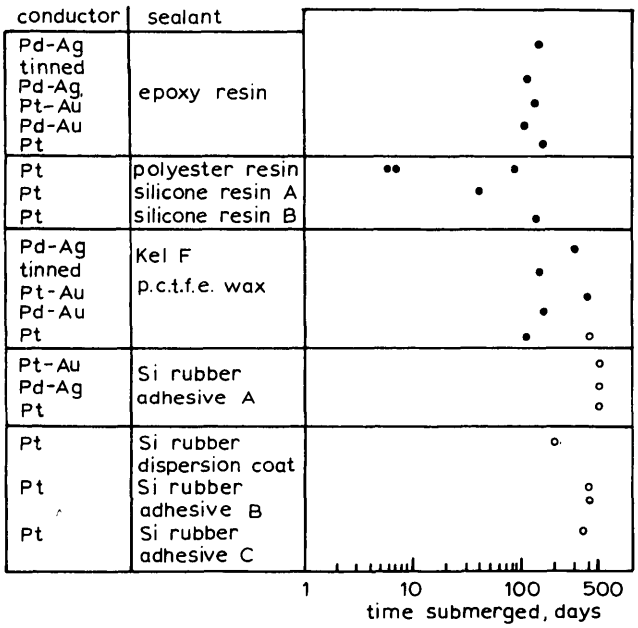

Fig. 14

Results of sealant tests when batch of three samples successfully reached 500 days

● *sample failed*
○ *sample satisfactory*

3.6 Search for a wire insulation

The choice of a suitable material for insulating the fine (75 μm diameter) platinum wires, with which we proposed to make the interpackage connections and the output cables, must lie among one of two main groups: the rubbery materials, such as silicon rubber or plasticised p.v.c., which are relatively thick and which rely for their resistance to mechanical damage on their ability to withstand hoop stress, and the enamels, which are thin, and whose resistance to damage is due to a sufficient hardness coupled with a firm bond to the underlying metal. Enamelled wires are evidently to be preferred, because they take up less room in the implant. The object of our wire tests was to find an enamel that was likely to stand up to the conditions of moisture and electric field present in the implant and, more severe, in the output cables, where the insulation would have to work wet.

In preliminary tests, we connected one end of a length of enamelled wire to one terminal of a variable d.c. power supply, by way of a milliammeter and limiting resistor. A bight in the wire 75 mm long was then pushed below the surface of some 1% saline in a small beaker. A large platinum plate, dipped into the saline, was connected to the other power-supply terminal. The power-supply voltage was set in the range 10-150 V and was turned on.

The experiment consisted merely in measuring how long it took the insulation to fail somewhere along the submerged portion of wire. The specimen was deemed to have failed when the fault current reached 1 mA. We could have chosen some different value as a criterion of failure and would have obtained much the same results, for the failures were of the catastrophic, rather than the progressive, type.

We began by testing an epoxy-based enamel which was alleged to have proved useful for electrophysiological work. Samples failed disappointingly quickly. Fig. 15 shows the dependence of mean times to failure on applied voltage for this enamel, both for wire anodal and wire cathodal to the bath. The points are geometric means of 15 determinations. The material was clearly useless for our high-voltage stimulation, although it was probably acceptable for biological recording. Since we were not interested in low-voltage performance, all further tests were carried out at 50 V using the same circuit as was used for testing sealants (Fig. 13), the test pattern being removed, of course, and the ends of the specimen being connected to points A and B and the bath to point C. If the insulation failed (or if the wire broke or became disconnected), the neon lamp would go out. Fig. 16 shows histograms of time to failure at 50 V for some enamels examined. Polyurethane enamel is of interest, because, when melted with the iron, it forms a flux which is helpful in soldering to the wire. As a submersible insulation, it did better

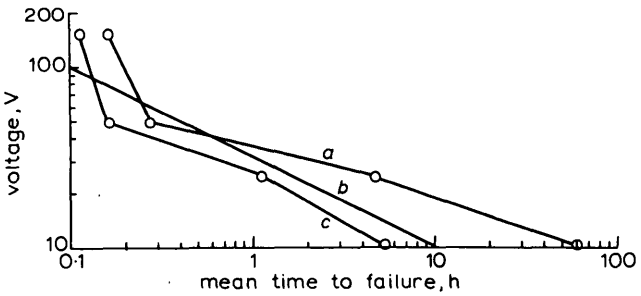

Fig. 15

Lifetimes of submerged samples of epoxy-enamel-coated platinum wire

The life varies, very approximately, inversely as the square of the applied voltage

a Wire cathodal
b $t = 1000/V^2$
c Wire anodal

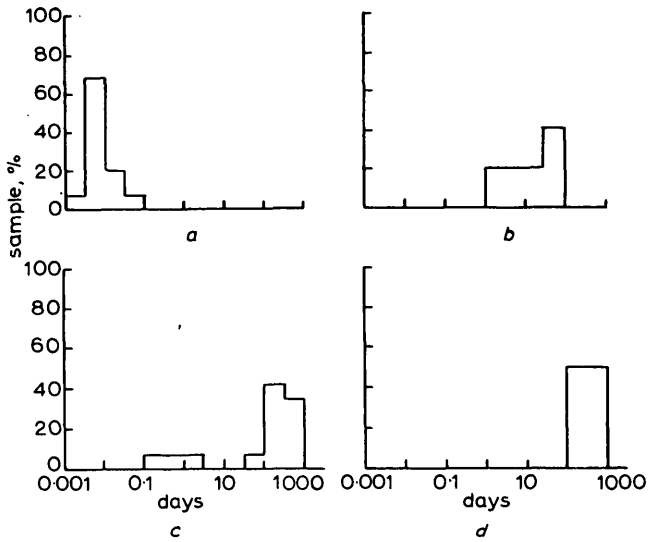

Fig. 16

Histograms of submerged lifetimes of samples of variously enamelled wires with a potential of 50 V across the insulation

a Epoxy-resin-platinum
b Polyurethane-copper
c Polyimide-platinum
d Oleoresin-copper

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* *Silicon rubber adhesive A was Midland Silicones Silastoseal E, an ideal material in almost all respects, now unfortunately no longer manufactured. Adhesive B was Dow Corning 3140, a material rather too stiff for convenience (viscosity 35 Pa s). Adhesive C was a blend of equal volumes to Dow Corning 3140 and Midland Silicones resin type 2628. The object of the added resin was to reduce the viscosity of adhesive B without interfering too much with its adhesion*

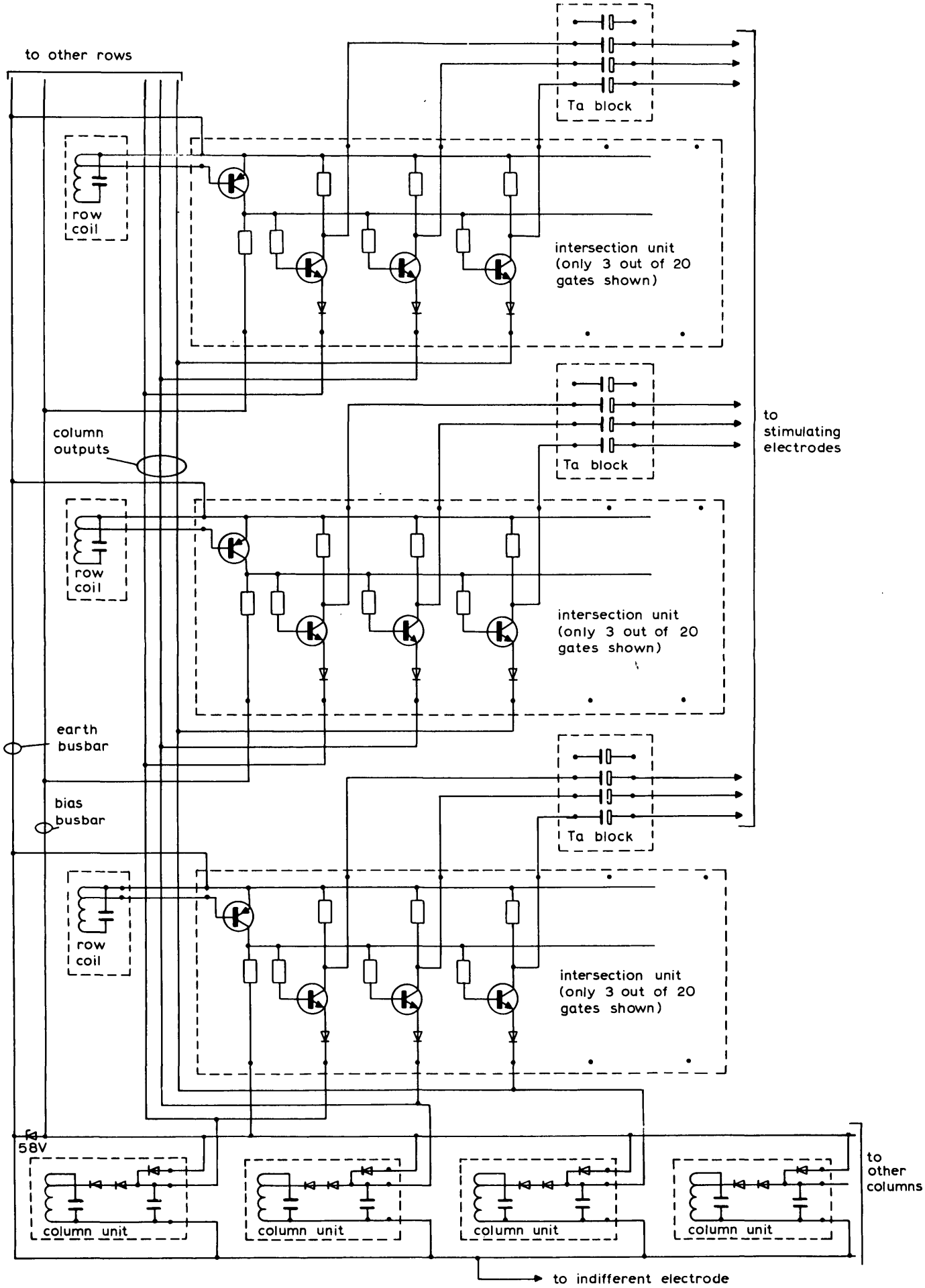

Fig. 17

Part circuit diagram of implant showing wiring for three rows and three columns

Evidently, more rows can be added, as shown, as well as up to 17 more columns

than the epoxy resin. Of the polyimide-insulated wire we tried, 'parts of it were excellent'. An American polyester enamel showed a bimodal failure distribution similar to that for polyimide. We found that oleoresinous enamel, which has been available for many years, performed better than any of the modern plastics. Nevertheless, we concluded that the enamels are probably unreliable and unpromising to examine any further; we turned to consider the rubbers.

Plasticised p.v.c. had proved to be satisfactory in the vapour-susceptibility tests, but is unsuitable for implants because it softens below autoclaving temperature. We were unable to find a commercial organisation which would extrude a silicon-rubber coating on to fine wires for us which would be sufficiently thin. Fortunately, G. Ettlinger drew our attention to a commercial p.t.f.e.-coated platinum wire.* The material is very expensive—about £5/m for 125 μm-diameter pure platinum coated with 25 μm of p.t.f.e., but it proved to be exactly what we needed; no sample of p.t.f.e.-insulated wire has ever failed. The p.t.f.e. can be prepared for bonding to sealants by etching it in commercial sodium-bearing preparations.†

3.7 Microelectronics packages

With the materials tests completed, it became possible to see how the total electronics circuit of the implant might be broken down into small packages, and how the packages would be encapsulated (Fig. 17). There would be five types:
(a) Column units (half tuned to 6 MHz, half tuned to 8 MHz), which would convert r.f. energy to direct current and direct it to one intersection in each row
(b) Intersection units, one to each row, containing all the AND-gate transistors for that row
(c) Row coils, tuned to 10 MHz, one for each intersection unit and, when r.f. energy is fed to it, opening all the gates in that intersection unit
(d) Tantalum capacitor blocks, a capacitor being included in the lead to each stimulating electrode to remove the d.c. component of the stimulating currents
(e) Zener diode to limit voltages present in the implant to 58.

Column units, row coils, the Zener diode and tantalum blocks would contain no vapour-susceptible components and could be sealed in silicon-rubber adhesive; the intersection units, since they contain planar transistor chips, would require vapour protection. The range of passivated transistors available at the time seemed both narrow and unsuitable (October 1969), and so we decided to use ordinary planar chips in a hermetic enclosure.

3.8 Manufacture of packages

3.8.1 Row coils

These were wound from polyurethane-insulated 0·15 mm-diameter copper wire on a temporary former. They were then bound with polyester sewing thread, removed from the former, dipped in epoxy-resin adhesive‡ and set aside to cure. After curing, the 100 pF ceramic-chip tuning capacitor was soldered across the ends of the winding (Fig. 18).

3.8.2 Tantalum capacitor blocks

Making these called only for cutting short the connecting leads of individual hermetically sealed tantalum capacitors, and gluing the capacitors into bundles as required by the final detailed design.§

3.8.3 Column units

The conductor pattern was printed on to 'as-fired' substrates‖ 0·64 mm (0·025 in nominal) thick, using a pure platinum ink.¶ Components were attached using silver-bearing solder. Column-unit coils were made, up to the cure stage, in a similar manner to the row coils (Fig. 19). The measured output resistance of a column unit was about 200 Ω away from resonance, rising to a much higher value in the range 1-2 kΩ at resonance, where the tuned-circuit e.m.f. is affected by the load.

3.8.4 Intersection units

A commercial organisation had developed for us a 10-gate intersection submodule, using 60 V transistors, on a substrate 10 mm square (Fig. 20). At that time, we did not know

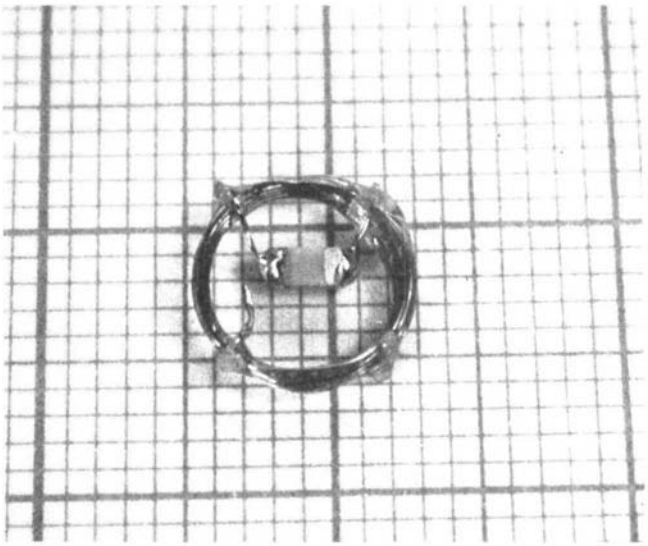

Fig. 18
Row coil and tuning capacitor
Small squares are of 1 mm side

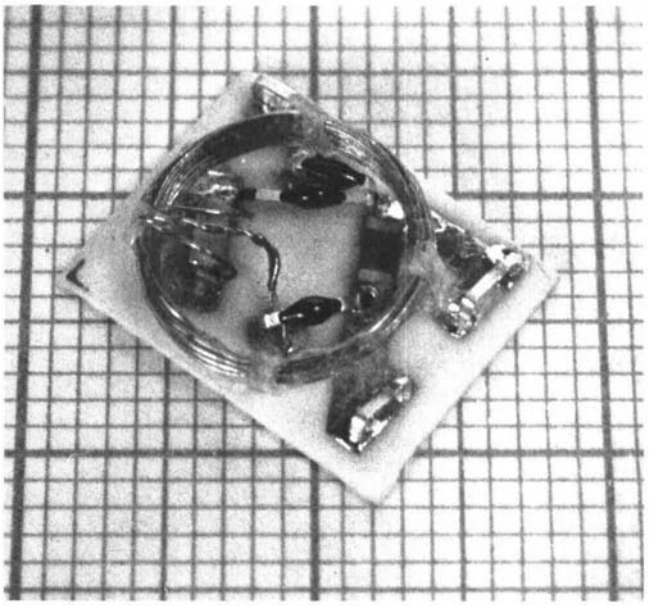

Fig. 19
Column unit
Small squares are of 1 mm side

how many rows and columns there would be in the eventual implant, but we were confident that the number of columns would lie between 10 and 20. We therefore needed a hermetic package that would just accommodate two submodules, the p-n-p row switching transistor and the common biasing resistor; it would need to have 43 leads. We could find nothing suitable commercially available, and were obliged to commission a special hermetic enclosure for this project (Fig. 21). It consists of a rectangular ceramic wall 1·27 mm high and 1·27 mm thick (0·050 in nominal) and of overall dimensions of 25·4 × 15·2 mm (nominal 1 × 0·6 in), soldered down—using a suitable glass—to an alumina base on which the 43 leads have previously been printed. The top of the wall is metallised, as is the appropriate area on the bottom of the lid. There is a small hole in the middle of the lid surrounded, on the top surface, by a circular metallised area.

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* *Made by the Medwire Corporation, USA*
† *Chemgrip, Tetraetch*
‡ *Araldite 752/956*
§ *Using Dow Corning Medical Adhesive type A*
‖ *Smith's substrates were used*
¶ *Johnson Matthey WKR 711*

The submodules are placed inside the package and wired to the lands using 25 μm gold wire. The lid is then attached using tin-lead solder, and the package tested. If all is well, the package is filled with a suitable inert gas and is sealed with a spot of solder over the hole in the lid. Normal industrial hermeticity tests then follow (Fig. 22).

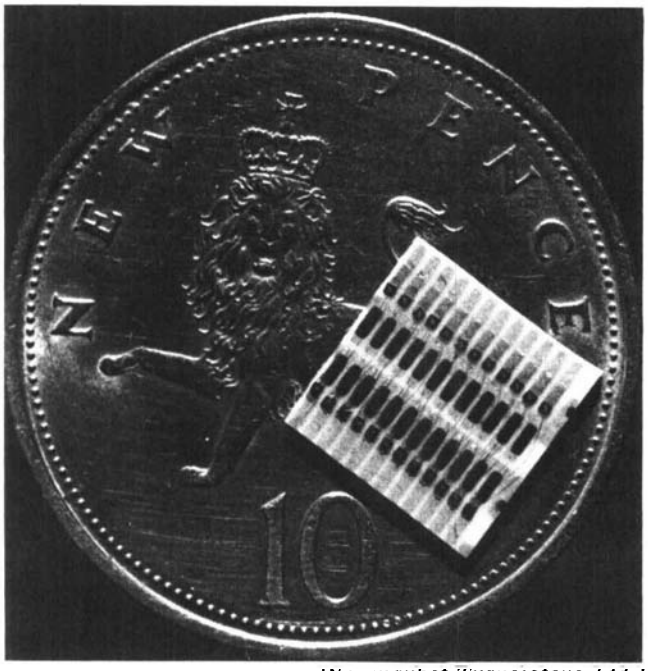

Fig. 20 *[Newmarket Transistors Ltd.]*

Thick-film 10-gate intersection submodule

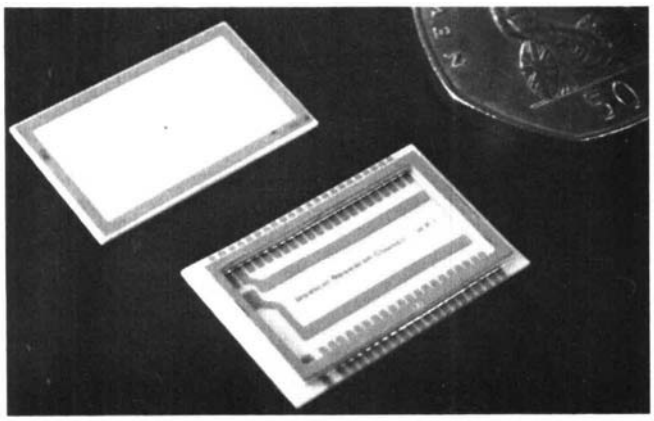

Fig. 21

Medical Research Council hermetic package, empty

Fig. 22

Medical Research Council hermetic package ready for use
Small squares are of 1 mm side

In general, the package follows a method of construction developed by J. Corkhill, of the RAE, Farnborough, England for use in British satellites,[9] but which had not previously been implemented in such a small size as was required for the prosthesis project. The success of the package is greatly to the credit of the companies who successfully made and used the walls.*

Later versions of the package, those used in the implant, were printed using the previously mentioned pure platinum ink. A variety of palladium-silvers and platinum-golds were used in earlier developmental models.

3.9 Submerged test circuits

Before finally embarking on the detailed design and construction of the implant, a last series of long-term tests was begun to check that the packages described in Section 3.8 would work correctly together, over a long period, in a simulation of the conditions inside an implant. The submerged test circuits (s.t.c.s) were really 'mini-implants', consisting of four column units, one intersection unit and one row coil. They were built, so far as possible, in exactly the same way as we proposed to build the implant.

A body is made in casting silicon rubber, having receptacles into which the packages fit. The packages are wired up using p.t.f.e.-covered platinum wire and silver-bearing solder (Fig. 23). The earth wire and the four outputs are led away to 1·5 kΩ dummy loads and to a 4-beam oscilloscope. The s.t.c. is tested using the implant driver (Section 4). There must be no output if a column unit is excited when the row coil is unexcited, or when it is excited asynchronously.

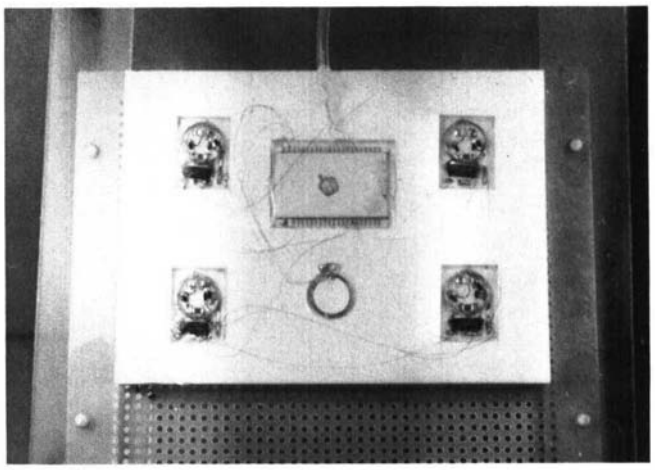

Fig. 23

Submerged-test circuit after wiring and before sealing

The outputs must be of satisfactory amplitude, and there must be no crosstalk between them. If the s.t.c. is working properly, the packages may be sealed, one at a time. The receptacle chosen for sealing is filled with chloroform, and the chloroform is drawn off a few seconds later and thrown away, being replaced with fresh chloroform; this is repeated once or twice more. The last lot of chloroform is thrown away, and a few minutes are allowed for the surface of the package to dry and for the receptacle to revert to its normal shape (the rubber will have swollen during the above defluxing and degreasing operation). The package is lifted clear of the receptacle, using clean forceps, and the receptacle is one-third filled with sealant. The package is then gently lowered into its receptacle again. When it has sunk to the bottom, more sealant is added, as necessary, to fill the receptacle to the brim.

When all packages have been sealed, a sufficient time should be left for the sealant to cure, and the wiring should be gently flattened and held flat by pressing down on it a piece of polyethylene terephthalate bobbin net† cut to overlap the

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* *Andermann & Ryder of Molesey, Surrey, and Newmarket Transistors Ltd., respectively*
† *John Small & Tidmas type 9/10¼*

body by 50 mm all round. The s.t.c. is placed on a sheet of balsa wood and the net is secured to the balsa wood with drawing pins, so that it lies tightly over the s.t.c. A paint should be made up.* By brushing on a succession of coats and letting each dry before beginning the next, the wiring can eventually be encapsulated in silicon rubber, the meshes of the net will be filled up, and the surface of the s.t.c. will become smooth. The drawing pins should be removed and the surplus netting cut away, taking care not to snip through any of the output leads.

We have made a total of 11 s.t.c.s following the foregoing procedure, or with apparently minor variations in it. S.T.C.s 1, 2, 4, 5 and 7 were sealed in epoxy resin, and although we improved our technique steadily, none of them worked properly for longer than six months. In nearly all cases, crosstalk developed eventually between the outputs, confirming for pulsed unidirectional fields in epoxy resin our evidence from the sealant tests using steady fields. S.T.C.s 3, 6, 8, 9, 10 and 11 were sealed in adhesive silicon rubber.† S.T.C. 3 was a disastrous failure, and showed crosstalk after only two days submerged; a possible explanation is that we coated the packages of this s.t.c. with a primer that was supposed to improve the adhesion of silicon-rubber sealants. S.T.C.s 6, 8, 9, 10 and 11 were made without any attempt at surface preparation other than chloroform washing, and have been successful. Fig. 24 shows s.t.c. 6, photographed on the day it had operated satisfactorily for a year, at 50°C, five days a week. The s.t.c. can be seen in its glass jar filled with saline, accompanied by a tubular fishtank heater. The five r.f. oscillators which control it are opposite the s.t.c., outside the jar.

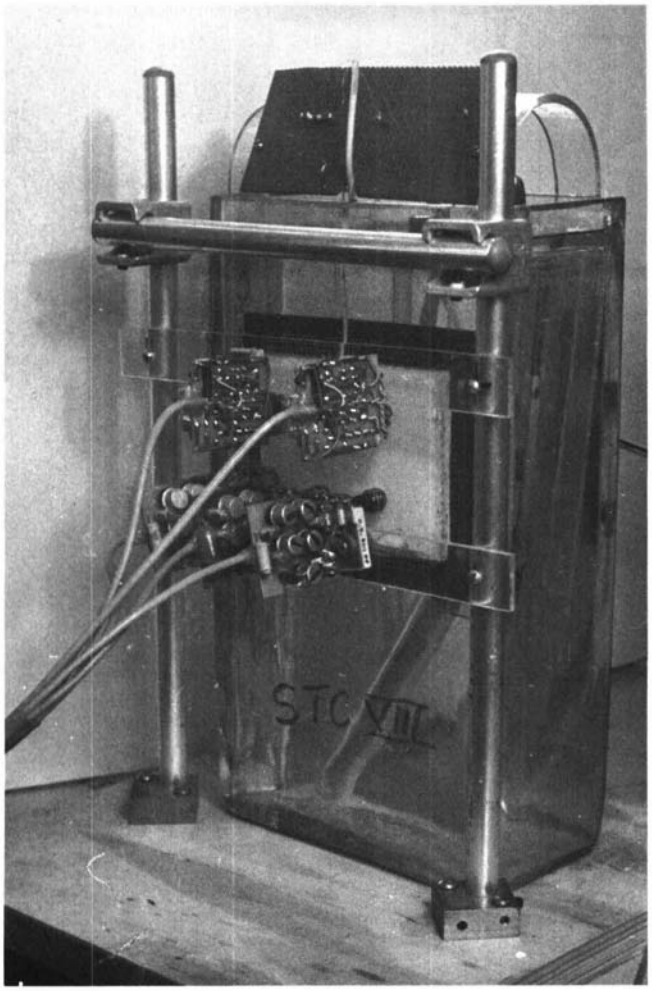

Fig. 24

Complete submerged-test circuit after one year of successful operation under saline

Note the five oscillators outside the glass tank, opposite the submerged coils visible in Fig. 23

* *Using equal volumes of Dow Corning Medical Adhesive type A and xylene*
† *Silastoseal E*

3.10 Implant toxicity

The successful s.t.c.s demonstrated that the problem we had expected to be the most difficult to solve—that of moisture penetration—seemed, in fact, to be soluble, and by a technique which, by a happy coincidence, also conferred autoclavability. We then had to find out whether it would be legitimate to implant a prosthesis made in the same way. The materials directly in contact with the host tissues are all known to be biocompatible, i.e. medical-grade castable silicon rubber, medical adhesive, platinum, p.t.f.e. and woven polyester fibres; we expected no difficulties from these surface materials. Inside the implant, there are alumina, glass, platinum, gold, epoxy resin, silicon (from which, again, we expected no difficulty), Kovar, copper, tin-lead solder and the industrial grade sealant.† Kovar, copper and solder are to be regarded as hazards, and it is essential that they should not escape. A typical implant would contain some 20 ml of the industrial sealant, and it was important to know whether significant amounts of water-extractable contaminant might leak into the surrounding tissue from this source.

In considering the metals, we carefully examined column units and intersection units which, sealed in the industrial sealant, had operated submerged for long periods. On carefully picking away the sealant, we never found any evidence of corrosion, but found that the surfaces were bright and that there was no perceptible staining of the surrounding sealant. We took this to mean that the metals were stable.

The industrial sealant was examined on our behalf by an independent analytical laboratory,‡ who looked for any low-molecular-weight rubber fractions or traces of the filler used leaking out of the material in the presence of saline solution at 38°C. The conclusions of the laboratory were that low-molecular-weight fractions of the silicon rubber were present, since they could be extracted with ether, but that they remained within the compass of the rubber in the presence of saline. As to the fillers, if a 50 g sheet of industrial sealant having a surface area of 500 cm^2 is immersed in 100 ml of saline for a week, the amount of titanium leached out will be less than 0·25 parts, and the amount of zinc will be less than 2·5 parts, per million parts by weight of the original rubber.

To find whether the continuous small loss of titanium and zinc, or any other material undetected by the laboratory, is biologically harmful, tests were carried out at the Institute of Psychiatry and by an independent laboratory.§ At the Institute of Psychiatry, a complete miniature implant, consisting of four column units, one row coil, one intersection unit and four tantalum capacitors, and using the industrial sealant for impregnating the wiring and the other netting (thus bringing the sealant in contact with tissue) was implanted extracranially into a baboon by Prof. Brindley. Three months later, this implant was removed; the tissue surrounding it was found to be perfectly healthy. At the independent laboratory,§ pieces of sealant were subjected to a cell-culture test, according to the method of Guess et al.;[10] they passed the test. Water extracts of sealant were injected into an anaesthetised cat to see if there was any observable effect on the blood pressure; there was none. Slivers of sealant were implanted into rabbit muscle; the animal was later killed and the implant sites were examined histologically. Here, the result was unexpected; the laboratory workers reported the presence of a fibroblastic inflammatory zone 0·1 mm thick around the implant, and categorised the reaction as 'severe toxic'. They commented that the cell-culture test is more stringent than the implant test, and that it is unusual for a material to pass the former and fail the latter.

Looking at all of the evidence relating to the sealant, bearing in mind the fact that we should not bring it into direct contact with tissue, and that it was the best sealant material we knew, we decided that it was right to use it in the human visual-prosthetic implant. On completing the implant, we should in any case wish to test it submerged for several weeks to ensure there were no gross deficiencies of manufacture. This testing period would also offer a useful opportunity for any water-extractable contaminants to leach out.

‡ *Yarsley Testing Laboratories Ltd.*
§ *Consultox Laboratories Ltd.*

3.11 Implant-body manufacture

The step from a submersible, autoclavable, implantable s.t.c. to the extracranial part of an actual implant requires only an increase in circuit complexity and a technique for casting the implant body. The body must fit the patient's skull, not just somewhere, but exactly where intended, so that the intracranial parts of the implant come into correct opposition with the occopital lobes. The thickness must be carefully controlled so that the cross-section is a slim lune. Besides the receptacles for the electronic packages, the body must be provided with stepped holes for the screws which hold it to the skull, and additional holes or slits through which tissue can eventually grow, providing extra stabilisation of the implant. We have worked out a technique, involving 13 steps, by which suitable bodies can be made:

(i) Place a rubber bathing cap on the patient's head (Fig. 25b)
(ii) Cover the bathing cap with plaster of Paris (Fig. 25c).
(iii) Lift the plaster of Paris and cap from the patient, allow the patient to depart and peel the cap from inside the plaster of Paris and discard it. The plaster of Paris shape remaining is the 'outer female' (Fig. 25d).
(iv) Smear the inside of the outer female with grease, and then use it to cast a plaster-of-Paris 'outer male' (Fig. 25e)

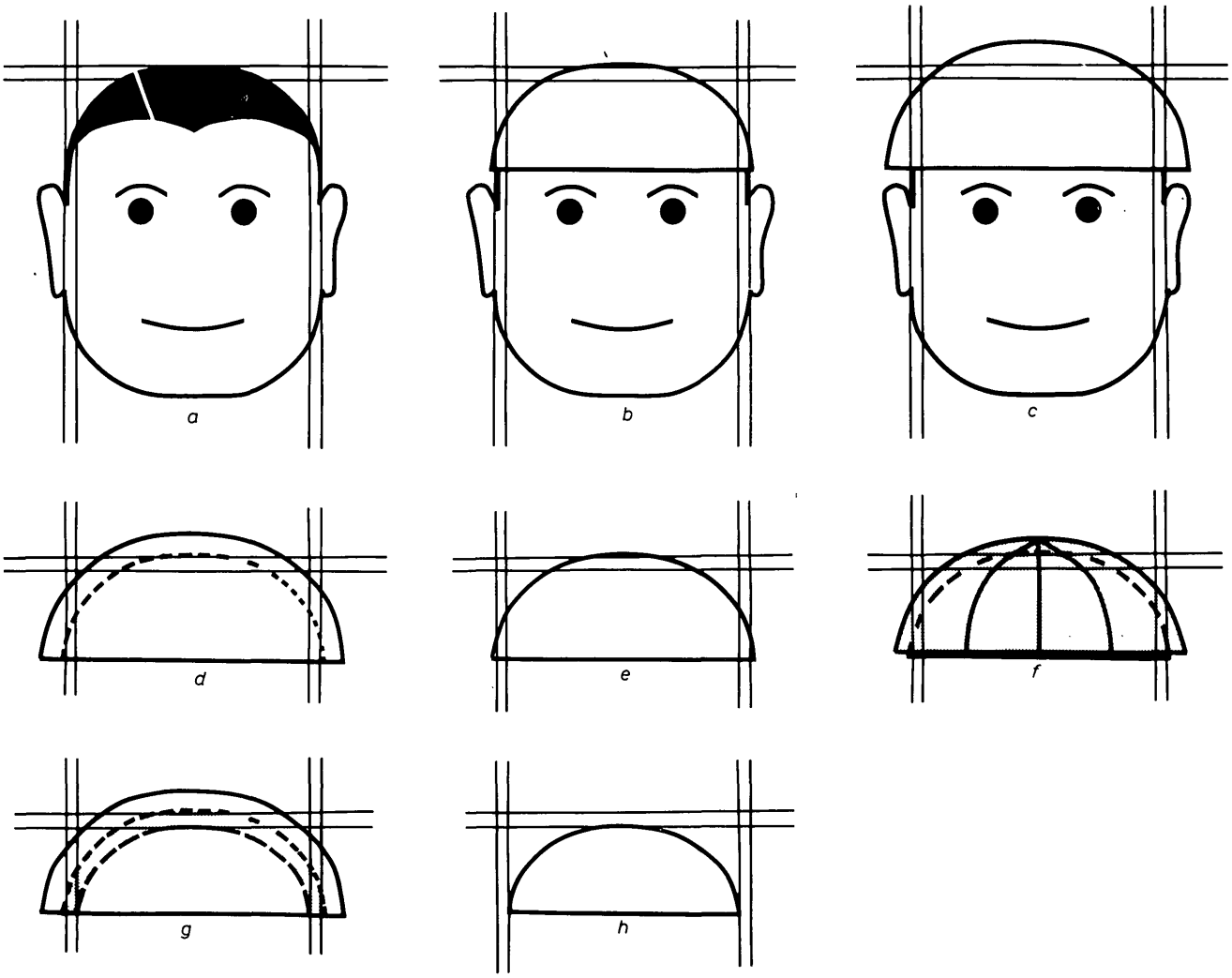

Fig. 25
First seven stages in production of an inplant body
Assume 6·4 mm thickness represents hair, scalp and bathing cap

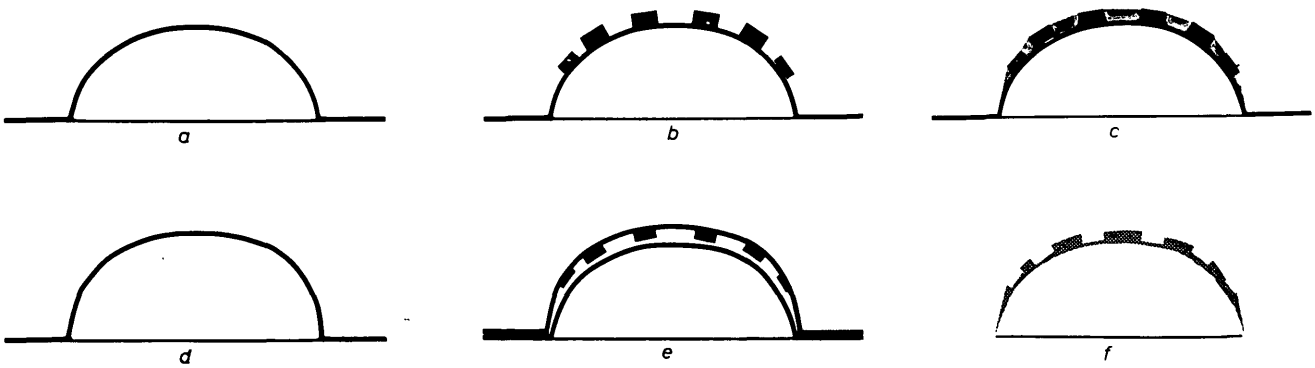

Fig. 26
Stages 8-13 in production of an implant body

(v) Make a 'schoolboy's cap' out of petal-shaped pieces of 6 mm-thick foam rubber glued together, so that it fits the outer male. It need not fit very well (Fig. 25f).

(vi) Put the schoolboy's cap into the outer female (Fig. 25g).

(vii) Use the new assembly to cast a plaster of Paris 'inner male'. If it is true to assume that the thickness of the foam rubber equals the thickness of the bathing cap plus the thickness of the patient's hair and scalp, the inner male should have the same shape as the patient's skull. If one has any reason to assume that the cap, hair or scalp are extraordinary in any way, the thickness of the foam rubber should, of course, be suitably changed. It would be wise, before proceeding, to check the contours of the inner male against X rays of the patient's skull (Fig. 25h).

(viii) Using a vacuum forming machine and 1 mm-thick p.v.c. sheet,* make a plastics shell from the inner male (Fig. 26a). This is the 'inner shell'.

(ix) Make up a number of rectangular Perspex blocks, 'thick blocks', one to represent each microelectronics package, of the same dimensions in plan as the plan dimensions of the eventual receptacle required, but thicker than the depth of the eventual receptacle by the thickness of rubber floor eventually wanted under the receptacle. We favour receptacle 3·16 mm (nominal $^1/_8$ in) deep, (since all our microcircuits are less than $^3/_{16}$ mm thick) and we like to have 1·58 mm ($^1/_{16}$ in nominal) of rubber floor beneath the receptacles; so we use $^3/_{16}$ in (4·8 mm) Perspex for these blocks. Plan the detailed layout of the implant by manoeuvring the blocks on the outer surface of the inner shell, attaching them temporarily with double-sided adhesive tape, until they form a compact group. Stepped cylindrical Perspex pieces, 'screw-hole moulds', representing the eventual positions of holes through the body for surgical fixing screws, are added at this stage. Keep at least 3 mm spacing between adjacent blocks, lest the walls between receptacles of the eventual body turn out to be too thin and weak. When satisfied with the layout, anchor each block permanently with a suitable glue or screw (Fig. 26b).

(x) Fill up the space between the blocks with plaster of Paris. Sufficient plaster must be used to come flush with the outer surfaces of the Perspex blocks, but no higher. At the edge of the group of blocks, sculpt the plaster of Paris carefully into a feather of triangular cross-section, so that the thickness of plaster of Paris falls away progressively towards the periphery (Fig. 26c).

(xi) Make a new plastics shell*—the 'outer shell'—from the assembly just completed. If stage (x) has been correctly carried out, the new shell should be faceted, having a precise rectangular facet for each Perspex thick block (Fig. 26d).

(xii) Now prepare a set of Perspex blocks, 'thin blocks', in plan view each of the same dimensions as its corresponding thick block, but of the same thickness as the depth of receptacle required (3·16 mm). Screw each thin block to the inner surface of the outer shell, each block locating in its corresponding facet (Fig. 26e).

(xiii) Use this assembly with another, new, inner shell as a 2-piece mould for the implant body. The two shells may be held in the correct relative position by spacing them with a fresh lot of cylindrical Perspex 'screw-hole moulds', and passing an 8BA machine screw through each spacer and through holes drilled in the outer and inner shells, securing it with a nut. Before pouring in the rubber† it is worth catching a piece of polyethylene-teraphthalate bobbin net between the two shells. This will become cast into the body and will constitute a useful reinforcement (Fig. 26f).

3.12 Detailed design and construction of the extracranial implant

When we became committed to the implant technology described in Section 3.7, we thought at first in terms of a unit having 20 columns and 9 rows. We made a body for such an implant, filled with receptacles with wax, stretched some netting over and painted it with silicon rubber. This dummy extracranial device was then handed to our neurosurgical collaborator, M. A. Falconer, for consideration. Mr. Falconer had no doubt at all that the dummy device was too big and that there would be severe difficulties at surgery in closing the scalp over it; would we please think again? We cut the number of columns from 20 to 15 and the number of rows from 9 to 5, and eliminated the tantalum capacitors in series with all but three of the outputs, relying on the capacitance-like properties of the electrode-tissue interface to keep the mean stimulating current zero. This is admissible for temporal patterns of stimulus within the range that we proposed to use. A dummy 15 × 5 extracranial device was handed to the surgeons on the 26th July 1971, for appraisal, and was accepted as satisfactory on the 2nd August. The body is shown in Fig. 27. Assembly of the implant began on the 6th September. The body was supported on the inner male mould, which was itself supported on a universal work clamp (Fig. 28). Arm and wrist rests helped greatly in making the small precise movements required, and all work was carried out under ×10 binocular magnification. Room lighting served for illumination, supplemented by a low-voltage filament benchlight. Soldering was carried out using silver-bearing solder and a 15 W soldering iron.

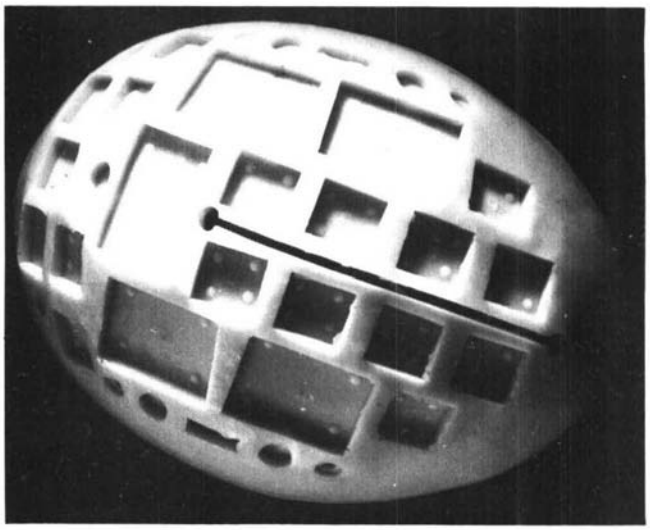

Fig. 27

Complete implant body

Receptacles for five row coils, five intersection units, 17 column units, two Zener diodes and six fixing screws can be seen

Note also the large centreline slot which allows some distortion of the body should it not otherwise fit quite properly

Fig. 28

Work station for building implant

* *Cobex was used in this case*

† *Dow Corning Medical Grade 382*

The soldering bit was the tip of a length of 24 s.w.g. tinned-copper wire wrapped around the point of the soldering iron. A variable transformer interposed between the public electricity supply and the iron was of great help in optimising the size and shape of the joints produced. As each intersection was wired up, the output lead was colour coded with small paint spots to identify it, and was then temporarily soldered to one input of a multi-input monitoring device, having a battery of circuits like that shown in Fig. 29, so that the presence of pulses in that output lead was indicated by the lighting up of the corresponding bulb. In this way, the satisfactory operation of the implant could be checked, at any time, as wiring up proceeded.

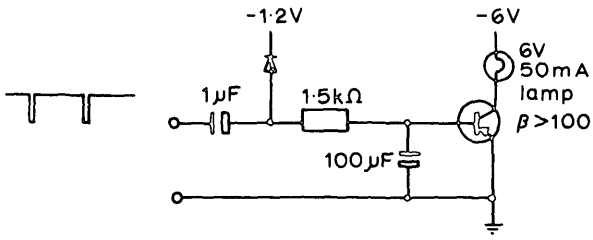

Fig. 29

One channel of implant-output monitor

One such circuit is needed for each output from implant

Wiring up was satisfactorily completed on the 4th October 1971, and the slow business of sealing the packages into their receptacles, one by one, and allowing each to cure before moving on to the next began. This took until the 5th November. Netting over and painting, as described in Section 3.9, was completed by the 17th November, when the implant was removed from the assembly rig and transferred to the submerged test rig (Fig. 30). It remained there, working satisfactorily until the 17th December, when it was removed and left to dry out.

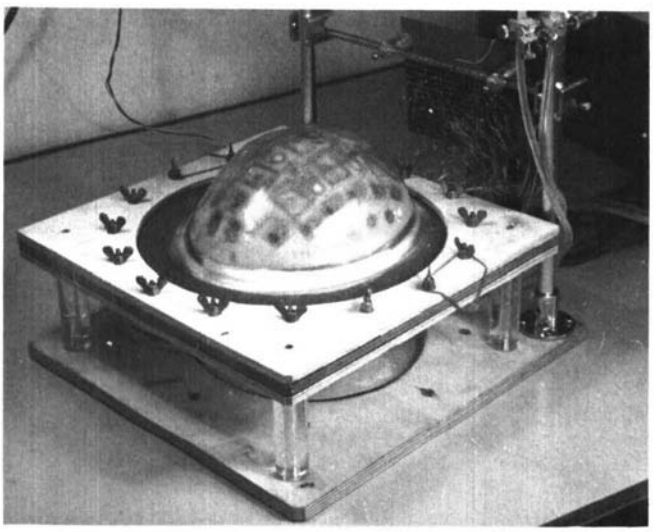

Fig. 30

Submerged-test rig

Implant is inside a plastics model of the top of the patient's head, which is filled with warm saline. It may be tested at at any time using oscillators held suitably outside the dummy head

3.13 Construction of intracranial implant

Compared with the labour of learning how to make, and then making, the extracranial part of the implant, the making of the two intracranial stimulating arrays was a simple matter. Two plaster-of-Paris models, one each of the left and right occipital poles of a typical human brain, were prepared. Each was mounted on a block of balsa wood 10 mm thick, so that the medial surface and most occipital regions were on the side of the model remote from the block. A release layer of thin polythene sheet was drawn tightly over each model and secured by pins pushed into the balsa. Pieces of polyethylene teraphthalate bobbin net were then also drawn over the models and similarly secured by pins (Fig. 31).

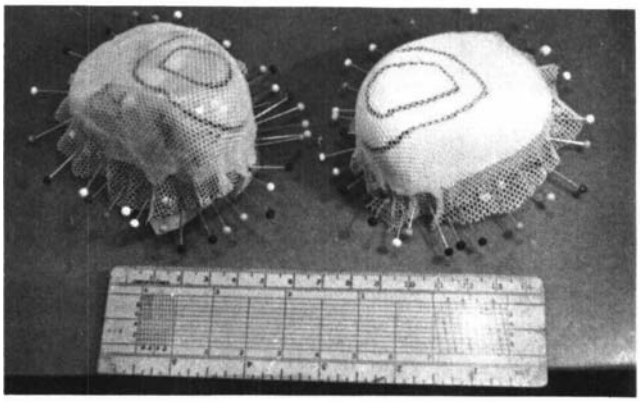

Fig. 31

Making intracranial electrode arrays: the polyethylene teraphthalate netting can be seen stretched over the plastics models of the left and right occipital poles

The extracranial implant was freed of its monitoring device by unsoldering the output cables, and the implant and one of the prepared brain models were screwed to a jig which held them in the correct relative position and enabled both to be offered up towards a microwelding head (Fig. 32). The electrodes were made from pieces of 50 μm thick (0·002 in nominal) pure platinum foil 4 × 1 mm. Fabrication of an electrode consisted in choosing any output lead from the bundle, noting the colour code, cutting the lead to a convenient length to reach to the point on the brain model where the electrode was to be, stripping the insulation back 1 mm with a microflame, welding a piece of platinum foil to the end of the wire, then hooking the foil through the polyethylene teraphthalate netting and bending up the ends to secure it (Fig. 33), being careful to make a note of the electrode position in the array.

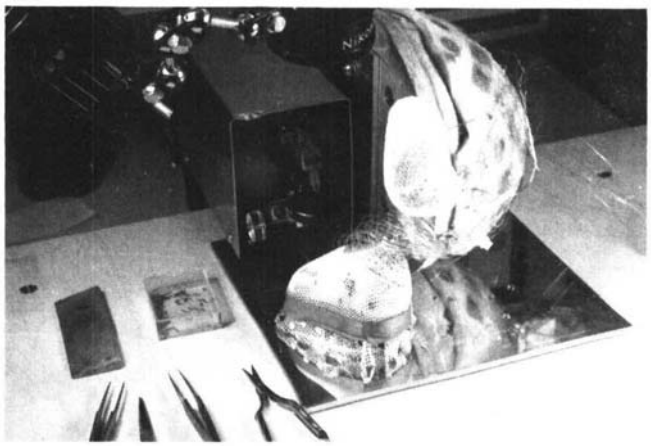

Fig. 32

Making intracranial electrode arrays: welding stimulating electrodes to output leads, then hooking them into the netting

The left-hand array has been completed and twisted into a most unphysiological position to allow the microscope to view work on the right-hand array

The total number of electrodes in each array is about 65. Of these, 37 or 38 are fed from the implant; the remainder are welded to wires which are lightly glued to the top of the extracranial implant but are not connected to anything. The object of having a total of 130 electrodes for an implant only capable of controlling 75 of them has to do with the possibility that one might, at some future date, wish to change this patient's extracranial implant for a better one. If it should

turn out that the intracranial part has grown in and cannot be removed, the patient has 130 electrodes available instead of 75.

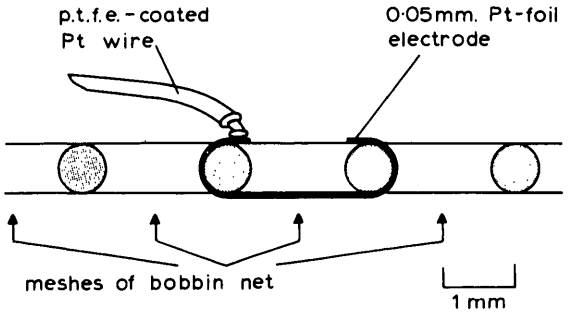

Fig. 33

Section through the polyethylene teraphthalate netting, showing how the stimulating electrodes are secured across the meshes

When all electrodes in the array were connected and in position, the wires were dressed neatly into small bundles side by side and secured with polyethylene-teraphthalate sewing thread and marline hitches. The aim here was to run the cable to each electrode array in the form of a flat tape rather than a single bundle of circular section, as, in this way, the cable would be less obtrusive after implantation. Silicon rubber paint* was then brushed through the wiring and down through the netting in among the electrodes and on to the release layer on the brain model. Sufficient layers were built up to fill the meshes of the net. More is inadvisable, as this paint contracts when the xylene evaporates, and there is great danger of the array distorting when it is removed from the mould if large masses of rubber are built up in this way. With the meshes of the netting filled, therefore, the array was completed by smoothing over a thick layer of casting rubber.† This did not shrink on curing, and gave 'body' and strength to the array, burying the wiring at the same time. When cure was complete, the array could be freed from the brain model by cutting carefully through the netting and impregnating rubbers with sharp scissors.

The other array was then made in a similar manner, the parts of the assembly jig being reversed to turn the jig into its mirror image.

Construction of the intracranial arrays was begun on the 20th December 1971 and completed on the 24th January 1972. The implant was by now practically complete. All

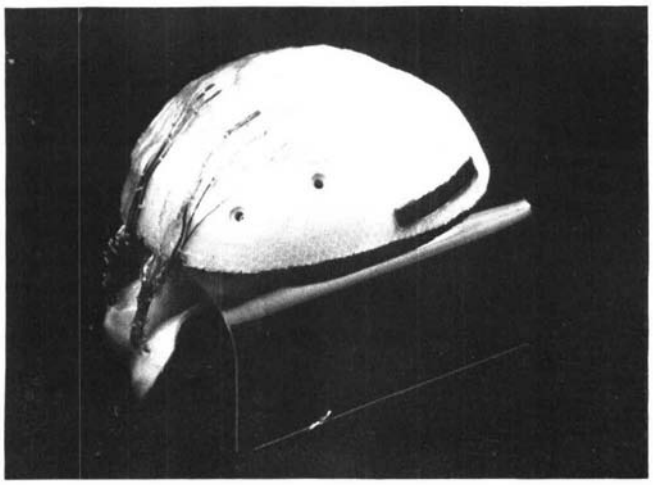

Fig. 34

Three-quarter view of completed implant

that remained was to add the two large platinum-indifferent electrodes. These were cut from 0·05 mm foil and welded to two wires specially provided for the purpose, which had been left emerging through the outer surface of the extracranial implant, and which were connected to the earth busbar inside. The indifferent electrodes were then degreased in chloroform and stuck down.‡ Figs 34 and 35 show the completed implant.

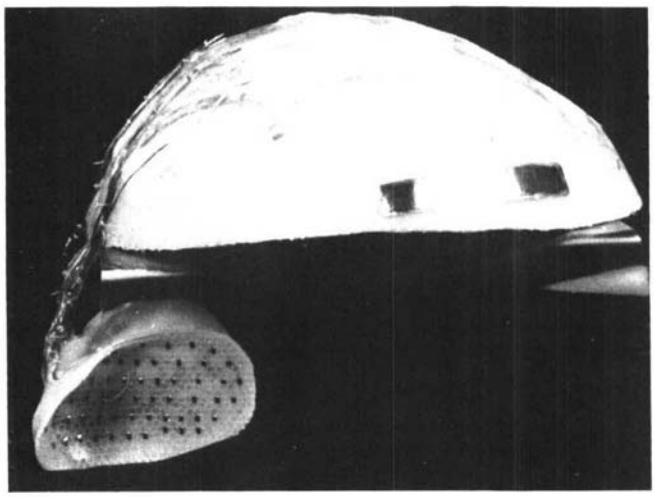

Fig. 35

Side elevation of completed implant

4 PRACTICAL DESIGN OF IMPLANT DRIVER

Fig. 36 shows a block diagram of the implant driver as built for the 20 column × 9 row implant originally planned. The oscillators are worn on the helmet and the remainder of the apparatus is carried on a wheeled rack. The change to a 15 × 5 implant means merely that row oscillators 6-9 and column oscillators P — T are omitted. No other modifications are needed.

A multivibrator produces a 0·5 ms pulse every 6 ms. These are fed to the row-1 power amplifier, to all control potentiometers in row 1, and to the first of a chain of halfshots. Each halfshot produces a 0·5 ms pulse whose leading edge coincides with the trailing edge of the pulse which feeds it. Thus a succession of pulses is available to turn on subsequent row oscillators, via their respective power amplifiers, in sequence ('scanning the rows'). As the ith row oscillator is switched on, a pulse also appears across all control potentimeters in row i. To show the patient phosphene B_3, for example, one closes switch contacts B_3, thereby causing column oscillator B to be turned on, via its power amplifier, for the same time as row oscillator 3 is turned on. The brightness of the phosphene is determined by the strength of the output of oscillator B, and may be preset by potentiometer B_3.

In the light of experience, we have modified the implant driver in three ways:

(a) We reduced the pulse duration to 0·2 ms.
(b) We increased the pulse period to 10 ms.
(c) We introduced an option whereby the output of the multivibrator (multivibrator 1) may be interrupted by the action of another multivibrator (multivibrator 2). Multivibrator 2 allows the system to run for 0·1 s, and then rest for 0·9 s etc. Such interrupted trains of pulses provide a more physiological type of stimulation for the patient.

The switches for controlling the phosphenes are, at first, manually operated toggle switches; this is a useful arrangement for presenting very simple patterns to the patient. In later work, they are changed to reed switches electromagnetically controlled from some other information source, such as a paper-tape reader or camera.

* *Made from diluted Medical Adhesive type A*
† *Dow Corning Medical Grade 382*

‡ *Using Medical Adhesive type A*

5 RESULTS

The prosthesis was implanted into our patient, a 64-year-old man who had been blind for over 30 years with retinitis pigmentosa, on the 4th February 1972. The patient fully understood what the implant was like and what might be expected from it. Indeed, he had followed its development keenly from the earliest days. Postoperative X rays are shown in Figs. 37 and 38. Testing of the implant was begun ten days later. A full account of the physiological and psychophysical aspects of these tests will be given elsewhere. In what follows, the emphasis will be on the implant performance from an engineering viewpoint.

Ten days after operation (day D + 10), when testing began, there was still a good deal of fluid under the patient's scalp, which meant that the shape of his head was different from its eventual form. There could therefore be no question of fitting and using a helmet to hold all the oscillators simultaneously in place. Further, an effect of the fluid was to separate abnormally the oscillator coils from their opposite numbers in the implant, reducing the strength of stimulus we could deliver. Again, we had to avoid holding oscillator coils against the patient's scalp in the neighbourhood of the incision and its stiches, which restricted the number of columns available to us. We therefore began by using two hand-held probe oscillators, synchronously pulsed, one for rows and one for columns. For checking objectively the

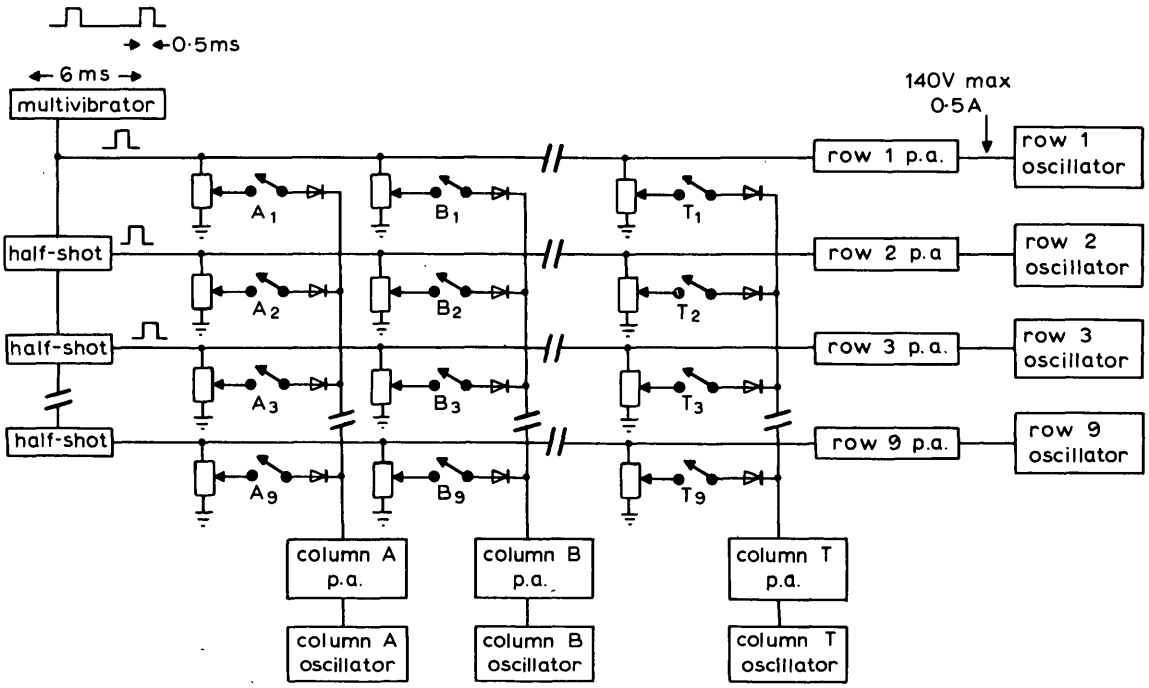

Fig. 36

Block diagram of implant driver

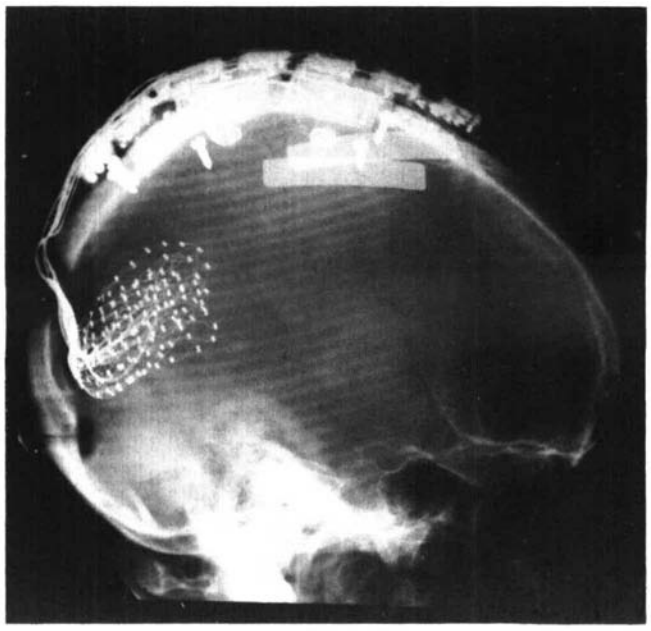

Fig. 37 *[M.A. Falconer and R.D. Hoare]*

Postoperative X ray, lateral view, showing extracranial part of prosthesis in position under scalp and stimulating electrode arrays over calcarine cortex

The right-hand array is angled slightly higher than the left-hand array

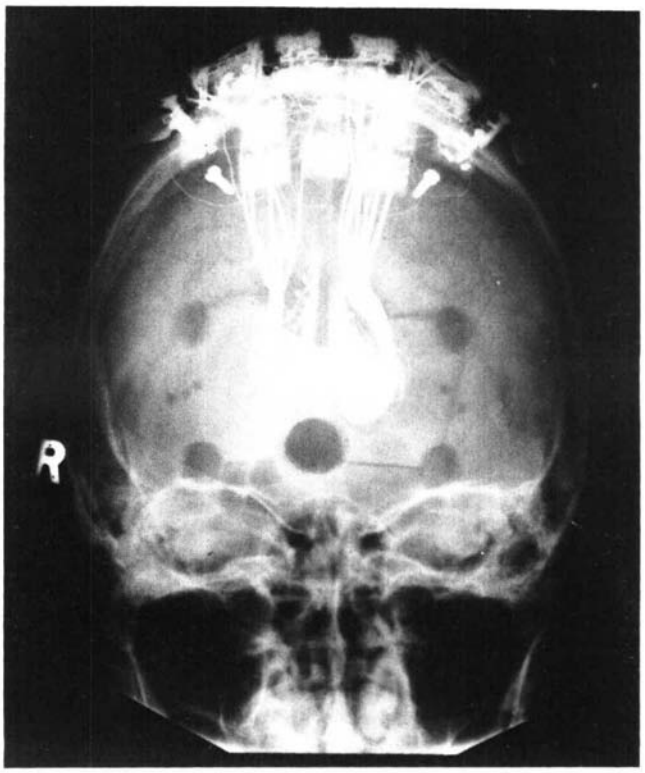

Fig. 38 *[M.A. Falconer and R.D. Hoare]*

Postoperative X ray

Posterior view showing prosthesis and occipital brow flap

operation of the implant, stick-on e.e.g. electrodes were attached to the patient at the forehead and at the back of the neck, and were connected to an oscilloscope. Delivery of a typical stimulating current by the implant causes a scalp potential of some tens of millivolts to be detected by the electrodes. By exciting one row and one column, this simple measurement provides a useful check that the relevant intersection is switching correctly, that the relevant conductor in the output cable is unbroken and that current is being delivered to the cortex. It gives no warning of crosstalk, but, in the event, we have not found this a disadvantage. Four rows and seven columns were checked that day. All 28 intersections produced scalp potentials, but only a few produced perceptions of light. The following day (day D + 11), we tested the whole implant. 65 out of 75 intersections produced correct scalp potentials, and of these perhaps a dozen gave phosphenes, albeit diffuse and not very bright. The 10 intersections which gave incorrect scalp potential were all intersections in column A (which would not turn off when row excitation was removed) and all intersections in column O (whose scalp potentials were very small). On the third day of testing (day D + 12), the number of phosphenes had risen to 33, and by day D + 13 it was 55.

With plenty of phosphenes available, Prof. Brindley, D.N. Rushton and the patient settled down to day after day of testing, during which the long job of mapping the positions in the visual field of the various phosphenes was undertaken. While this was going on, the implant underwent various vicissitudes. On day D + 20, all intersections in column N abruptly ceased to give scalp potentials; on day D + 74, when the patient returned from a holiday, intersections in column F were found to have failed also. On the other hand, the fluid under the patient's scalp had, by then, subsided, and intersections in column O were now working normally; we measured the patient for his helmet two days later. While the helmet was being assembled, the powerful valve oscillators which had been built for exciting the first implant were used to look for, and successfully find, some more high-threshold phosphenes, thereby raising the score of intersections which had produced phosphenes to 74 out of 75. By day D + 84, the helmet was completed and operating successfully, and with perhaps the best 60 phosphenes at the experimenters' disposal, all mapped as to position, the presentation of patterns to the patient, using the array of toggle switches on the implant driver, could be begun.

The results here were rather disappointing. It was becoming clear that phosphenes seen by our second patient were larger than those of our first and that, when pairs were elicited simultaneously, they tended to fuse together to form a single, bigger phosphene. On the other hand, our new patient was very good at distinguishing between phosphenes presented sequentially, even when the phosphenes overlapped. In a typical experiment, we chose six good phosphenes (i.e. bright and clearly defined at the edge) which lay approximately in two vertical columns of three, the format for Braille characters. We then tried to present the patient with English text, one letter at a time, each letter being converted into its Braille form as some combination of the six phosphenes. Presenting the phosphenes simultaneously, he scored 56% of letters correct, whereas presenting the phosphenes sequentially, dwelling about 100 ms on each phosphene, he scored 79% correct.

On day D + 120, the condition of the implant was found to be substantially unchanged: columns F and N were out of action, intersections in column A were now switching off rather better than they had done earlier, and all other columns worked normally. To demonstrate the prosthesis in action, we made the apparatus shown in Fig. 39. English text was punched on to 5-hole tape, using the standard teleprinter code. The patient was provided with two switches. Switch A advanced the tape through the reader by one row of holes. The new letter was electrically translated into Braille form, and, on the patient pressing switch B, the information was presented sequentially to him in 600 ms. With this equipment, he achieved a reading rate of 7 letters/min and scores in excess of 90% with very little practice. The six phosphenes chosen for this work did not form a very accurate pair of columns, and the patient commented that most of the time taken to read a letter was lost trying to identify the distorted Braille pattern with which he was provided, rather than in actually perceiving the pattern; he was quite confident that he could speed up with practice.

Our patient demonstrated his ability to 'read' from paper tapes in this way at the Oxford meeting of the Physiological Society on the 3rd June 1972.[11]

Fig. 39

Patient reading Braille symbols presented singly

In the centre of the photograph is the patient wearing the helmet and holding two switches. On the left is Dr. Rushton scoring and timing the patient's performance. At the extreme bottom right is the 5-hole-tape reader and in the extreme right foreground is the equipment for translating from teleprinter code to Braille code. Behind this, immediately on the patient's left, is the implant driver

6 DISCUSSION

When this work was begun, the most severe engineering problem to be overcome appeared to be the protection from the effects of moisture, in the presence of appreciable electric fields, of a complex microelectronics system. While it would be premature to claim that we have been successful in this, we believe the prospects to be good; our experience with adhesive silicon-rubber sealants is that devices protected in this way, if they do not fail in the first few weeks after immersion in the moist environment, do not fail at all. No doubt, the high vapour-transmission rate of silicon rubber ensures that the implant quickly arrives at an equilibrium with its surroundings, after which very little more happens.

On the other hand, the faults which have, in fact, affected the implant, namely failure of interconnections, were quite unforeseen. We have observed similar interconnection failures in implants screwed to the skulls of baboons, but we had supposed that an implant in a young baboon received much more severe battering than did an implant in a 64-year-old man. From our knowledge of baboon-implant failures, we suspect that the cause of loss of columns F and N is failure of the adhesion of the printed conductor under either the earth or the output terminals of the column units. Similarly, the intermittent loss of switching-off capacity in column A is, we believe, caused by intermittent loss of bias consequent on adhesion failure of the ink under the bias terminal of column unit A; if one simultaneously excites a row coil, column A and some other column (say F), intersections in column A are then found to switch normally, bias being supplied from column unit F. If we had used a conductor ink possessing greater adhesion, and if we had coiled the connecting wires into helixes so that some strain could be accommodated before excessive tension was applied to the package terminals, we should, perhaps, have had 75 outputs available instead of 65.

However, what is wanted from future visual-prosthetic implants is not so much 75 outputs as 750. There is little doubt that, using the technology evolved for the second implant, there is not much hope of taking the number of outputs from a row-and-column prosthesis beyond 100, if the size is to remain acceptable. The best hope for obtaining more outputs, without the implant becoming larger, seems, to us, to lie in nonhermetically sealed intersection units using nitrided beam-lead chips, and we are now working toward an implant made in this way, having 304 outputs.

We do not believe that the disappointingly large phosphenes seen by our second patient have to do with the stimulus parameters or electrode size used, but rather that it has to do with the very long time (30 years) for which he has been blind and his youth (13 years old) when he began to go blind. It is possible that future visual prostheses will be of greatest help to those who have not been visually handicapped for very long.

7 CONCLUSIONS

The development and construction of an experimental visual prosthesis has been described; it enables a patient to read by seeing Braille letters presented singly. The results give grounds for confidence that better prostheses, of use to the patient, can be built using existing technology.

8 ACKNOWLEDGMENTS

To carry out the work described in this paper, the author was obliged to make himself familiar with, and become adept in, microelectronics techniques. Some of this can be learned from books and journals, but much of it exists as practical 'know-how' on the shop floor of manufacturing industry. I would like to acknowledge the immense amount of help given to this project by electronics engineers in many British firms, whether there was any likelihood of our becoming customers or not, once they had heard what we were trying to do. Particular thanks are due to the special products division, Newmarket Transistors Ltd., who made our intersection submodules and sealed them into our hermetic package.

It is a pleasure also to acknowledge the technical help given me by M. Rodel in the construction of the implant driver, and by E. Sayer in the work leading to our implant technology and in the construction of the implant itself. My thanks also are due to Prof. G. S. Brindley for extremely close collaboration throughout the project, and to the Medical Research Council of Great Britain for generous financial support.

10 REFERENCES

1 LEONARD, J. A.: 'Mobility and the blind—a survey', Med. & Biol. Eng., 1963, **1**, pp. 465-479
2 MELEN, R. D., and MEINDL, J. D.: 'Electrocutaneous stimulation in a reading aid for the blind', IEEE Trans., 1971, **BME-18**, pp. 1-3
3 BRUGLER, J. S.: 'Technology for the Optacon, a reading aid for the blind'. Proceedings of Eurocon 1971 (IEEE, 1971)
4 BACH-Y-RITA, P.: 'A tactile vision substitution system based on sensory plasticity' in STIRLING, T. (Ed.): 'Visual prosthesis' (Academic Press, 1971)
5 BRINDLEY, G. S., and LEWIN, W. S.; 'The sensations produced by electrical stimulation of the visual cortex', J. Physiol., 1968, **196**, pp. 479-493
6 MARG, E., FORDEMWALT, J. N., and MINER, J.: 'Design for a phosphene visual prosthesis', Brain Res., 1970, **19**, pp. 502-510
7 BRINDLEY, G. S.: 'Sensations produced by electrical stimulation of the occipital poles of the cerebral hemispheres, and their use in constructing visual prostheses', Ann. R. Coll. Surg., 1970, **47**, pp. 106-108
8 'Artificial vision', Wireless World, 1971, **77**, pp. 215-217
9 FISHER, H. D., and CORKHILL, J. R.: 'Advanced packaging for the Black Arrow X3 satellite'. Proceedings of the Internepcon conference, Brighton, England, 1969
10 GUESS, W. L., ROSENBLUTH, S. A., SCHMIDT, B., and AUTIAN, J.: 'Agar diffusion method for toxicity screening of plastics on cultured cell monolayers', J. Pharm. Sci., 1965, **54**, pp. 1545-1547
11 BRINDLEY, G. S., DONALDSON, P. E. K., FALCONER, M. A., and RUSHTON, D. N., 'The extent of the region of occipital cortex that when stimulated gives phosphenes fixed in the visual field', J. Physiol, 1972, **225**, pp. 57-58